

Analysis Notes 131AH

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If you are reading this...

Hello! These are my notes for analysis, which I have typed up to save to github so I can reference them in the future. Things will be written in my own words and may not be fully correct, but this is just my attempt to fully internalize everything and write it down as precisely as possible.

Contents

1	Propositional and First-Order Logic	2
2	Naive Set Theory and Zermelo Fraenkel	4
2.1	Naive Set Theory	4
2.2	Zermelo Fraenkel	4
3	Ordered Pairs and Relations	6
4	Isomorphisms of Dedekind Cuts and Ordered Fields	7
5	Properties of the Reals	9
6	Countability and Cardinality	15
7	Uncountable Sets	21
8	Metric Spaces	26
8.1	Metrics	27
9	Topology	32
9.1	Open and Closed sets (In Metric Spaces)	32
9.2	Topologies	33
9.3	More Topological Notions	38
9.4	Sequential Closedness	41
10	Metric Completeness	44
10.1	Key Statements and Definition of Completeness	44
10.2	Completeness Related to Closedness	45

1 Propositional and First-Order Logic

Proposition

Definition 1.1. A proposition is a statement that takes a TRUE or FALSE value

For example, "birds are mammals" is a valid proposition, which takes on the value of FALSE. We also define a "primitive proposition", which is one with no connectives or quantifiers.

Connective

Definition 1.2. A connective is a unary or binary operator allowing us to chain propositions to create new propositions

The connectives we will work with are $\neg$ (logical not), $\wedge$ (and), $\vee$ (or), $\implies$ (implies), and $\iff$ (biconditional). The $\neg$ operator is the only unary operator, which switches the value of the proposition. The others are binary, requiring two propositions and outputting one value.

We view an interesting truth table for $\implies$:

P	Q	$P \implies Q$
T	T	T
T	F	F
F	T	T
F	F	T

When P is false but Q is true, the entire statement is "vacuously" true as the expected proposition Q is true no matter P . When both are false, the proposition is true as well. So in general when P is false, $P \implies Q$ is true.

Proof by Contradiction

Lemma 1.3. $(P \implies Q) \iff \neg(P \wedge \neg Q) \iff \neg P \vee Q$

Here the biconditional means that the propositions are equivalent. We can use the biconditional to represent when two statements imply each other or are logically the same.

This lemma highlights proof by contradiction, where we assume P and $\neg Q$ and we show some contradiction (or show that the proposition is false). Then the logical not of the proposition is true. The second equivalence is an application of DeMorgan's Laws for logic.

NOTE: $(P \implies Q)$ is not the same as $\neg P \implies \neg Q$. It is equivalent to $Q \implies P$ by Lemma 1.2. We see the truth table below and compare it to the one above:

P	Q	$\neg P \implies \neg Q$
T	T	T
T	F	T
F	T	F
F	F	T

Proof by Contrapositive

Lemma 1.4. $(P \implies Q) \iff \neg Q \implies \neg P$

This is the method of proof by contrapositive. This implication is called the **"only if"** direction, while the reverse is the **"if"** direction.

Quantifiers

Definition 1.5. We propose two quantifiers: $\forall$ (for all/for each) and $\exists$ (there exists)

A first order statement assumes the form: (quantifier)(variable): (proposition). This is the basis of First Order Logic, an extension of propositional logic. For example:

$$\forall x : P(x)$$

translates to: for all x such that $P(x)$ is true. The terminology "some" is equivalent to $\exists$.

Quantifier Distribution

Lemma 1.6. $\neg(\forall x : P(x)) \iff \exists x : \neg P(x)$ and $\neg(\exists x : P(x)) \iff \forall x : \neg P(x)$

Here we see how the logical not operator distributes over quantifier and proposition.

Same Quantifier Swap

Theorem 1.7. $\forall x \forall y : P(x, y) \iff \forall y, \forall x : P(x, y)$ and $\exists x, \exists y : P(x, y) \iff \exists y, \exists x : P(x, y)$

If the quantifiers for two variables are the same, we can switch the order of the quantifiers. This will appear in proofs involving sets and supremums/infimums. However, if they are not the same, then we cannot suggest an equivalence.

Different Quantifier Swap

Theorem 1.8. $\exists x \forall y : P(x, y) \implies \forall y \exists x : P(x, y)$

The reason why this is a one-way implication is because the left hand statement is more specific. We will encounter this when dealing with images and preimages. We take the following example:

Mathematical Example: If there exists one x greater than all y , then for each y there exists an $x > y$. In this case that x is the same. For the converse, if each y has an $x > y$, there may not be a singular x greater than all y .

Figurative Example: If there is one building to which all apartments belong, then all the apartments belong to that building. However, if all the apartments belong to a building, not necessarily they belong to one building. We can have the apartments in multiple buildings.

2 Naive Set Theory and Zermelo Fraenkel

2.1 Naive Set Theory

Naive Set Theory was the initial proposed set theory. A set being a container for some sort of objects. To construct a set, we have the following:

Comprehension Principle

Definition 2.1. The Comprehension Principle: $A := \{x : P(x)\}$

The symbol $:=$ means "defined as". If we use $=$ when referring to sets, that is not the same as using $:=$.

The Comprehension Principle tells us that a set is some elements x that satisfy a proposition $P(x)$. We won't dwell on this theory too much, but it has pitfalls such as Russell's paradox:

$$A := \{x : x \notin x\}$$

Here if $x \notin x$, then we could say $A \notin A$, but the proposition tells us that $A \in A$, so we come across a contradiction.

2.2 Zermelo Fraenkel

Axiom of Separation

Axiom 2.2. Any set A can be defined as $\{x \in U : P(x)\}$.

We see that the elements $x \in A$ are being drawn from U which in this case can be a universe (all the possible sets, but this is not itself a set), or a bigger set. By restricting elements, we can prevent self references such as those in Russell's paradox.

Axiom of Extensionality

Axiom 2.3. $\forall A, B : A = B \iff \forall x : (x \in A \iff x \in B)$

We mentioned previously that $:=$ is not the same as $=$ for sets. Extensionality gives the definition of $=$ to be that the sets must have the same elements. The axiom itself is sometimes written as an implication rather than a biconditional.

Axiom of Empty Set

Axiom 2.4. $\exists \emptyset : \forall x \in U : x \notin \emptyset$

We postulate the existence of the empty set, which contains none of the possible elements in the universe. Now, how do we build the set of all subsets (aka the powerset)?

Subset Relation

Definition 2.5. $\forall A, B : A \subseteq B \implies \forall x \in A : x \in B$

Let's try to use the Axiom of Separation and this definition to construct the powerset. We can say maybe for a set A that $\mathcal{P}(A) := \{B \in C : B \subseteq A\}$. But this definition requires a set C to exist in which B belongs,

so either this is the powerset or a set containing more elements than the powerset, and so we have a circular definition.

Powerset

Axiom 2.6. $\forall A \exists \mathcal{P}(A) : (\forall B \subset A : B \in (\mathcal{P})(A))$

We have the same issue when we go to define a set such as $\{x, y\}$, which is called the pairset, as each element is also inherently a set.

Pairset

Axiom 2.7. $\forall x, y \exists A \forall Z \in A : x = z \vee y = z$

The pairset allows us to show the existence of a singleton set $\{x\}$ if both the elements given are the same as $\{x, x\} = \{x\}$. This can be shown by extensionality.

Infinite Set

Axiom 2.8. $I := \{x : x \in I \implies \{x\} \in I\}$

This is one possible construction of the infinite set we will use for the naturals; however, this axiom postulates the existence of infinite sets. Many of the constructions in analysis will hinge on this axiom; without it, we fail to construct the idea of "natural numbers" let alone "real numbers".

3 Ordered Pairs and Relations

We talk about pairs (or n-tuples) of numbers such as $(1, 2)$ or (x, y) . However, we also distinguish the order unlike sets, saying that $(1, 2)$ is not the same as $(2, 1)$.

Kuratowski Pair

Definition 3.1. $(x, y) := \{\{x\}, \{x, y\}\}$

4 Isomorphisms of Dedekind Cuts and Ordered Fields

So far we have constructed the reals or $\mathbb{R}$ as the set of Dedekind cuts, which are the set of rationals less than that real number. By allowing for completeness, we can show that each cut admits an upper bound so it admits a supremum.

The supremum is the real number itself; however, quantifying the supremum will require the existence of an algebraic structure that encompasses the supremum. If we solely rely on $\mathbb{Q}$, then certain cuts may not admit supremums in $\mathbb{Q}$.

So we say there exists some complete ordered field $(F, +, 0, \cdot, 1, \leq)$ (completeness is important here, otherwise we cannot guarantee supremums). Each ordered field fundamentally has the $\mathbb{N}_F$. We examine a theorem:

Ordered Field Characteristic

Theorem 4.1. If F is an ordered field, then it has characteristic 0

We prove this by contrapositive:

Proof. Assume F does not have characteristic 0. Then $\sum_{k=1}^n (1) = 0$ for an arbitrary n . We then know that $0 < 1$, by ordered field properties:

$$0 < 1 \implies 0 < \sum_{k=1}^{n-1} (1) < \sum_{k=1}^{n-1} (1) + 1 = \sum_{k=1}^n (1)$$

Now assume the inequality to be true, then by multiplicative property of ordered fields:

$$\implies \sum_{k=1}^{n-1} (1) \cdot \sum_{k=1}^{n-1} (1) < \sum_{k=1}^n (1) \cdot \sum_{k=1}^{n-1} (1)$$

$$\iff (n-1)^2 = n^2 - 2n + 1 < n(n-1) = n^2 - n \quad (1)$$

We know that n or 1 added n times repeatedly is equal to zero, so we get:

$$0 + 1 = 1 < 0$$

Which is a contradiction of a key property of ordered fields that $0 \leq 1$. □

So now, the addition of 1 to an element of the field becomes our successor operation, and 0 becomes our zero element. If we then consider the subset of elements greater than zero, then zero is not a successor of any number in the set. Thus we can construct the naturals in a field as:

$$\mathbb{N}_F = \bigcap_{\alpha \in I} \{A_\alpha \subseteq F : (\forall x \in A_\alpha : x + 1 \in A_\alpha)\}$$

Now that we have the naturals in every ordered field, by field properties, we require their additive and multiplicative inverses. Furthermore any combination of these must also remain in the field, giving rise to the rationals $\mathbb{Q}_F$ (considered the "minimal" ordered field).

So currently we have the set of Dedekind cuts $\mathbb{R}$ and an ordered field structure F . One is constructed from ground up, the other's existence is assumed respectively. We aim to show the two are isomorphic. We know the following theorem proved in prior chapters:

Isomorphism of Rationals

Theorem 4.2. The rationals of different fields are isomorphic

So there exists an isomorphism we call $\psi : \mathbb{Q} \rightarrow \mathbb{Q}_F$. Since the Dedekind cuts are merely sets of rationals, then we can construct cuts in F which we call $\mathbb{R}_F$. We can extend $\psi : \mathbb{R} \rightarrow \mathbb{R}_F$.

The Supremum Isomorphism

Theorem 4.3. $\forall A \in \mathbb{R}_F : (\exists b \in F, \forall a \in A : a \leq b) : \sup : A \rightarrow F$ and $\sup$ is an isomorphism

This lemma is essentially the restatement of the completeness of F . If we take any subset bounded above, then it admits a supremum. However, here we define the supremum as a map from a set to an element in the field. We prove it is an isomorphism for a cut specifically. We require the lemma:

Dedekind Cuts in an Ordered Complete Field

Lemma 4.4. $\forall A \in \mathbb{R}_F : A = \{a \in \mathbb{Q}_F : a < \sup(A)\}$

For now we only show the inclusion $A \subseteq \{a \in \mathbb{Q}_F : a < \sup(A)\}$.

Proof. If $x \in A$, then $x \leq \sup(A)$. If $x = \sup(A) \implies \sup(A) \in A$. By the definition of Dedekind cut's, there cannot be a maximal element, so:

$$\exists b \in A : \sup(A) < b \wedge b \leq \sup(A)$$

The second inequality comes from the supremum definition. We see that we get a contradiction, so $\sup(A) \notin A$, thus $\forall a \in A : a < \sup(A)$. $\square$

We now return to proving the theorem at hand.

Proof. For the sake of brevity we assume $\sup$ is a homomorphism (which means that it respects the operations of sets and the field). We aim to show injectivity:

$$\forall A, B : \sup(A) = \sup(B) \implies A = B$$

If two cuts are not equal, then either $A \subset B \vee B \subset A$. Assume without loss of generality that $A \subset B$. Then:

$$\exists b \in B, \forall a \in A : a < b$$

This inequality we can derive from the ordering of F and the construction of the cuts. Then b is an upper bound for A and by definition of supremum $\sup(A) \leq b$. So by the lemma $a < \sup(A) \leq b < \sup(B)$. Thus $\sup(A) < \sup(B)$, proving injectivity of $\sup$. $\square$

Proof. We now prove the surjectivity of $\sup$:

$$\forall x \in F, \exists A \in \mathbb{R}_F : \sup(A) = x$$

By the total ordering of F and the fact that each field has its copy of the rationals, we can find the set of rationals strictly less than x . This is the associated Dedekind cut A . $\square$

Isomorphism of All Complete Ordered Fields

Theorem 4.5. For all complete ordered fields $(F, +, 0, \cdot, 1, \leq)$, we find that $\mathbb{R} \cong F$.

The proof of this is simply a combination of the previous theorems and lemmas. We know that $\psi : \mathbb{R} \rightarrow \mathbb{R}_F$ is an isomorphism and $\sup : \mathbb{R}_F \rightarrow F$ is also an isomorphism. The composition is therefore an isomorphism:

$$\sup \circ \psi : \mathbb{R} \rightarrow F$$

So the Dedekind cut representation of the reals are isomorphic to any complete ordered field. So now instead of dealing with the reals as a set of Dedekind cuts, we can deal with it as an ordered field of the supremums of the Dedekind cuts. So $\sqrt{2}$ is a supremum of the cut, so we can explicitly refer to the cut as $\sqrt{2}$ itself even though the cut does not admit a supremum in $\mathbb{Q}$.

5 Properties of the Reals

We have determined that every complete ordered field is isomorphic to the real numbers. We now aim to show certain properties have correctly carried forward to this construction.

We dealt with rational roots and equalities such as $x^2 = 2$ that did not admit solutions in the rationals, but allowed us to construct $\sqrt{2}$ in the reals. We now see the reals hold for arbitrary roots as well:

Existence and Uniqueness of the n th root

Theorem 5.1. $\forall a \in \mathbb{R}^+, \forall n \in \mathbb{N} \setminus \{0\} : (\exists x \in \mathbb{R}^+ : x^n = a) \wedge (x^n = y^n \implies x = y)$

In essence, there exists an n th root for every real number, which is unique. We use the following property:

$$\forall x, y \in \mathbb{R}, \forall n \in \mathbb{N} \setminus \{0\} : x^n - y^n = (x - y) \sum_{k=0}^{n-1} x^k y^{n-k-1}$$

Integer exponentiation can be defined by field properties as repeated multiplication of the element in the field.

HW 5 Problem: We prove the property below:

Proof. By the distributive property of multiplication in fields:

$$\begin{aligned} (x - y) \sum_{k=0}^{n-1} x^k y^{n-k-1} &= x \sum_{k=0}^{n-1} x^k y^{n-k-1} - y \sum_{k=0}^{n-1} x^k y^{n-k-1} \\ &= \sum_{k=0}^{n-1} x^{k+1} y^{n-k-1} - \sum_{k=0}^{n-1} x^k y^{n-k} \end{aligned}$$

We can notice a pattern of cancellations here. The last term of the left term is x^n and the first term of the right term is y^n . We can then rewrite the sum as:

$$= x^n - y^n + \sum_{k=0}^{n-2} x^{k+1} y^{n-k-1} - \sum_{k=1}^{n-1} x^k y^{n-k}$$

We see that the two summations output the same terms, just that the starting and ending indices are up by 1. If our starting term is xy^{n-1} , then at $k = 0$ we sum $x^{k+1}y^{n-k-1}$, but at $k = 1$, we sum $x^k y^{n-k}$.

If our last term is $x^{n-1}y$, then the same applies. So the two summations are equal, thus their subtraction equals 0, and our resultant expression is $x^n - y^n$. $\square$

We return to the proof of the theorem. We define the set $A := \{y \in \mathbb{R}^+ : y^n \leq a\}$. Since this is a subset of the field, if it is bounded, then it admits a supremum:

$$1 + a > a \implies (1 + a)^n > (1 + a)^{n-1}a > a^n > a \quad (\text{Ordering Property of Multiplication})$$

Thus $\forall y \in A : y \leq 1 + a$ so A is bounded above, thus it admits a supremum. We want to show that $\sup(A)^n = a$, thus defining the supremum of the set as the n -th root and showing its existence (as we know supremum exists).

Proof. We use the fact that $\forall y \in A : y \leq \sup(A)$, We first show $\sup(A)^n \leq a$. We also use the fact that:

$$\forall m \in \mathbb{N} \setminus \{0\} : \sup(A) - \frac{1}{m} < \sup(A) \implies \exists y \in A : \sup(A) - \frac{1}{m} \leq y$$

$$\sup(A)^n - a \leq \sup(A)^n - y^n = (\sup(A) - y) \sum_{k=0}^{n-1} \sup(A)^k y^{n-k-1}$$

Our previous implication that $\sup(A) - \frac{1}{m} \leq y \iff \sup(A) \leq y + \frac{1}{m} \implies \sup(A) \leq y$. Since $y \in A$, we know that $y \leq \sup(A)$, so $y = \sup(A)$. Our inequality becomes:

$$\sup(A)^n - a \leq \sup(A)^n - y^n = (\sup(A) - y) \sum_{k=0}^{n-1} \sup(A)^k y^{n-k-1} = 0 \iff \sup(A)^n \leq a$$

We now prove that $\sup(A)^n \geq a$. We note that $\forall m \in \mathbb{N} \setminus \{0\} : \sup(A) + \frac{1}{m} > \sup(A)$, so:

$$\begin{aligned} (\sup(A) + \frac{1}{m})^n > a &\implies a - \sup(A)^n < (\sup(A) + \frac{1}{m})^n - \sup(A)^n \\ &= \frac{1}{m} \sum_{k=0}^{n-1} (\sup(A) + \frac{1}{m})^k \sup(A)^{n-k-1} \leq \frac{n}{m} (\sup(A) + \frac{1}{m})^{n-1} \end{aligned}$$

Thus, by dividing both sides (also the Archimedian Property is really a double implication), we see that:

$$\frac{m}{n} \frac{a - \sup(A)^n}{(\sup(A) + \frac{1}{m})^{n-1}} < 1 \implies a - \sup(A)^n \leq 0 \iff \sup(A)^n \geq a$$

So we have proven that $\sup(A)^n = a$, proving the existence of the n th root, which we will denote by $a^{1/n}$. $\square$

Proof. We now prove the uniqueness of the n -th root. Assume there are two n -th roots of a such that $y^n = a = \tilde{y}^n$. Then:

$$\sup(A)^n = y^n = \tilde{y}^n = a \implies a^{1/n} = \sup(A) = y = \tilde{y}$$

Since the supremum is unique, then if $y = \sup(A) = \tilde{y}$, then y must equal $\tilde{y}$, so the n th root is unique. $\square$

Now that we have the notion of natural powers and n th roots, we can extend this to rational roots. We know that any rational can be represented as the ratio of two integers p/q . Note the following:

Integer Powers

Corollary 5.2. $\forall n \in \mathbb{N}, \forall a \in \mathbb{R} : a^{-n} = (a^{-1})^n = (\frac{1}{a})^n \wedge a^0 = 1$

With this corollary, we are able to define the notion of integer powers (as integers have additive inverses not accounted for by natural powers). Another property of natural powers and by extension integer powers:

Integer Powers

Corollary 5.3. $\forall m, n \in \mathbb{N} : a^m \cdot a^n = a^{m+n} \wedge a^{m \cdot n} = (a^m)^n$

HW 5 Problem: Using this corollary, we can prove the following lemma. For this problem assume $a > 1$:

Rational Powers Equivalence

Lemma 5.4. $\forall m, n, p, q \in \mathbb{Z} : n, q > 0 \wedge \frac{m}{n} = \frac{p}{q} \implies \forall a > 1 : (a^m)^{1/n} = (a^p)^{1/q}$

Proof. If $\frac{m}{n} = \frac{p}{q}$, then we know that $mq = np$ where the two are integers. If $m/n < 0$, then $p/n < 0$ and same for $m/n \geq 0$. Thus the two fractions take on the same sign. Therefore $mq = np$ will either be positive or negative.

If $mq = np < 0$, we simply define their additive inverses $m'q' = n'p' > 0$ and use them to construct an equality (eg $-a = -b \iff a = b$ by field properties). So without loss of generality we proceed with $mq = np > 0$:

$$a^{mq} = a^{qm} = (a^q)^m \wedge a^{np} = a^{pn} = (a^p)^n \quad (2)$$

This is true by properties of exponentiation with respect to naturals and field multiplication. Thus, by the existence of arbitrary roots:

$$(a^q)^m = (a^p)^n \implies ((a^q)^m)^{1/n} = a^p \implies ((a^m)^q)^{1/n} = a^p$$

We need to show that we can exchange the places of q and $1/n$, so we prove $(a^m)^{1/n} = (a^{1/n})^m : n > 0$:

$$((a^m)^{1/n})^n = a^m \wedge ((a^{1/n})^m)^n = (a^{1/n})^{nm} = (a^{1/n})^{nm} = ((a^{1/n})^n)^m = a^m$$

Since both expressions to raised to the $n : n > 0$ are equal, then the n th roots are also equal so $(a^m)^{1/n} = (a^{1/n})^m$.

So returning to our initial statement:

$$((a^m)^q)^{1/n} = a^p \implies ((a^m)^{1/n})^q = a^p \implies (a^m)^{1/n} = (a^p)^{1/q}$$

□

Rational Powers nth Roots

Corollary 5.5. $\forall p, q \in \mathbb{Z}, q \neq 0 : a^{\frac{p}{q}} = (a^p)^{1/q} = (a^{1/q})^p$

Rational Powers Addition

Lemma 5.6. $\forall r, s \in \mathbb{Q} : a^{r+s} = a^r \cdot a^s$

Proof. We let $r = \frac{m}{n}$ and $s = \frac{p}{q}$, where $m, n, p, q \in \mathbb{Z} \wedge p, q \neq 0$, then $r + s = \frac{m}{n} + \frac{p}{q} = \frac{mq+np}{pq}$. Thus by corollary 5.3:

$$a^{r+s} = a^{(mq+np)/pq} = (a^{1/pq})^{mq+np}$$

Now $mq + np$ is an integer expression, by corollary 5.2, we can rewrite the expression as:

$$= (a^{1/pq})^{mq} \cdot (a^{1/pq})^n = (a^{mq/pq}) \cdot (a^{np/pq}) = a^{m/p} \cdot a^{n/q} = a^r \cdot a^s$$

□

The idea of multiplicative inverses can now be brought about for rational powers. Let $p, q, m, n \in \mathbb{Z} : q, n \neq 0$, then:

$$a^{p/q} \cdot a^{m/n} = (a^{1/pq})^{mq+np} = 1$$

Using corollary 5.1, we account for the sign of $mq + np$. If the expression is greater than zero, then it is a natural number. If it is less than zero, we define its additive inverse $(mq + np)' > 0$ by field properties and change the base from a to a^{-1} as per the corollary.

Without loss of generality we proceed with the first case. Once again by corollary 5.1 stating that $a^0 = 1$:

$$mq + np = 0 \implies np = -mq$$

So by Lemma 5.2, $(a^{1/pq})^{mq+np} = (a^{1/pq})^{mq} \cdot (a^{1/pq})^{np} = (a^{1/pq})^{mq} \cdot (a^{1/pq})^{-mq}$. Since $mq \in \mathbb{Z}$, we again apply corollary 5.1 and 5.3:

$$(a^{1/pq})^{mq} \cdot (a^{1/pq})^{-mq} = a^{m/p} \cdot a^{-m/p} = 1$$

So we see that $(a^{m/p})^{-1} = a^{-m/p} = (a^{-1})^{m/p}$.

Now that we have defined and proven the properties of rational powers, we need to define real powers, which we do via supremums again leveraging completeness of the underlying ordered field:

Real Powers

Definition 5.7. $a^x := \begin{cases} \sup\{a^z : z \in \mathbb{Q} \wedge z \leq x\}, & a > 1 \\ \inf\{a^z : z \in \mathbb{Q} \wedge z \leq x\}, & a < 1 \\ 1, & a = 1 \end{cases}$

Proof. We prove the first case, where if $a > 1$ and $x \in \mathbb{Q}$, then $a^x = \sup\{a^z : z \in \mathbb{Q} \wedge z \leq x\}$. We know that $x < x + 1$. By density of the rationals:

$$\exists r \in \mathbb{Q} : x < r < x + 1$$

Since $z \leq x$, then $z < r$. We now need to prove the lemma:

Lemma

Lemma 5.8. $\forall z, r \in \mathbb{Q} : z < r \implies a^z < a^r$.

If $z < r$, and $a > 1$, we see that $a^{r-z} > a > 1$ so by Lemma 5.2:

$$a^{r-z} = a^r \cdot a^{-z} = a^r \cdot (a^z)^{-1} > 1$$

Since $a > 1 > 0$, then $a^z > a > 0$. So if $(a^z)^{-1} < 0$. Thus:

$$a^r \cdot (a^z)^{-1} > 1 \implies 1 \cdot a^z < a^r \cdot (a^z)^{-1} \cdot a^z \iff a^z < a^r$$

Therefore a^r is an upper bound, allowing for the set to admit a supremum. We now want to show that a^x is the least upper bound. Consider any upper bound $a^p : p \in \mathbb{Q}$, then:

$$\forall z \leq x : a^z \leq a^p \implies a^x \leq a^p$$

If $a^p < a^x$, then $a^{x-p} > 1$. Since $a > 1$, then $a^{x-p} > a > 1$ only if $x > p$.

$$p < x \wedge z \leq x \implies \exists z \leq x : p < z \implies a^p < a^z$$

Thus, a^p is not an upper bound for all a^z , so we have proven the statement by contrapositive.

Real Powers Addition

Lemma 5.9. $\forall x, y \in \mathbb{R} : a^{x+y} = a^x \cdot a^y$

We define $a^{x+y} = \sup\{a^z : z \in \mathbb{Q} \wedge z \leq x + y\}$.

Since $a^y = \sup\{a^p : p \in \mathbb{Q} \wedge p \leq y\}$ and $a^x = \sup\{a^q : q \in \mathbb{Q} \wedge q \leq x\}$:

$$a^x \cdot a^y = \sup\{a^q\} \cdot \sup\{a^p\} = \sup\{a^q \cdot a^p\} = \sup\{a^{q+p}\}$$

This follows from Lemma 5.2 and sup being a homomorphism. Multiplying the supremums combines the predicates in a logical $\wedge$ and results in a Minkowski product $\{a^{q+p} : q \leq x \wedge p \leq y\}$. By ordered field properties:

$$q \leq x \wedge p \leq y \implies q + p \leq x + y$$

By definition of the set of a^{x+y} , we see $\forall q \leq x, \forall p \leq y : a^{p+q} \in \{a^z : z \leq x+y\}$. Therefore:

$$\{a^{q+p}\} \subseteq \{a^z\} \implies \sup\{a^{q+p}\} \leq \sup\{a^z\} \iff a^x \cdot a^y \leq a^{x+y}$$

Thus $a^x \cdot a^y = \sup\{a^{q+p} : p \leq y \wedge q \leq x\} \leq \sup\{a^z : z \leq x+y\} = a^{x+y}$.

We now prove the opposite inequality, that $a^{x+y} \leq a^x \cdot a^y$. We choose a fixed $\tilde{q} \leq x$ and a fixed $\tilde{p} \leq y$, then:

$$a^{x+y} = \sup\{a^z : z \leq x+y\}$$

For every z , we can pick $\tilde{p}, \tilde{q}$ such that by Lemma 5.3:

$$(\forall z \in x+y : \exists \tilde{p} \leq y, \tilde{q} \leq x) : z \leq \tilde{p} + \tilde{q} \implies a^z \leq a^{\tilde{p}+\tilde{q}} = a^{\tilde{p}} \cdot a^{\tilde{q}}$$

Which means that since the choices of $\tilde{p}, \tilde{q}$ were arbitrary:

$$\forall z \leq x+y : a^z \leq a^{\tilde{p}} \cdot a^{\tilde{q}} \leq a^x \cdot a^y$$

By definition of supremum, a^{x+y} is the least upper bound for all a^z and we have shown that $a^x \cdot a^y$ is an upper bound, so:

$$a^z \leq a^x \cdot a^y \implies a^{x+y} \leq a^x \cdot a^y$$

Thus proving $a^{x+y} = a^x \cdot a^y$. This forces a multiplicative inverse to exist such that $a^x \cdot a^y = a^0 = 1 \implies x+y=0 \implies y=-x$. Thus $(a^x)^{-1} = a^{-x}$. $\square$

Now that we have defined exponentiation with respect to a real valued base, along with elementary operations, we aim to define the inverse of exponentiation, which is the logarithm.

The Exponential Function

Theorem 5.10. The exponential function: $f : \mathbb{R} \rightarrow \mathbb{R}^+ \setminus \{0\}$ is a bijection

We define $f(x) = b^x$ for any $a \in \mathbb{R}^+ \setminus \{1\}$. The proof is the same if $b > 1$ or $b < 1$. Then:

Proof. f is injective if $\forall x, y \in \mathbb{R}^+ : b^x = b^y \implies x = y$. We can use Lemma 5.4 and multiplicative inverses:

$$b^x = b^y \implies b^x \cdot (b^y)^{-1} = 1 \implies b^x \cdot b^{-y} = b^{x-y} = 1$$

Since $b > 1$, then by the definition of exponentiation, $x - y = 0$, so $x = y$.

If f is surjective, then $\forall y \in \mathbb{R}^+. \exists x \in \mathbb{R} : b^x = y$. Here x is called the logarithm of y base b .

We fix $a > 1$, then by the results of problem 1:

$$a^n - 1 = (a - 1) \sum_{k=0}^{n-1} a^k \cdot 1^{n-k-1} = (a - 1) \sum_{k=0}^{n-1} a^k \geq \sum_{k=0}^{n-1} 1^k = n(a - 1)$$

Now if we let a be the n th root of b such that $a = b^{1/n} > 1$, then:

$$a^n - 1 \geq n(a - 1) \implies b - 1 \geq n(b^{1/n} - 1)$$

If $t > 1$ and $n > (b-1)/(t-1)$, then $b^{1/n} < t$:

$$n(b^{1/n} - 1) \leq b - 1 < n(t - 1) \implies b^{1/n} - 1 < t - 1 \implies b^{1/n} < t$$

If w is such that $b^w < y$, then $b^{w+(1/n)} < y$ for sufficiently large n ; to see this, apply part (c) to $t = y \cdot b^{-w}$.

$$b^{1/n} < t \implies b^{1/n} < y \cdot b^{-w} \implies b^{(1/n)+w} < y$$

If $b^w > y$, then $b^{w-(1/n)} > y$ for sufficiently large n . Let $t = y^{-1} \cdot b^w > 1$:

$$b^{1/n} < t \implies b^{1/n} < y^{-1} \cdot b^w \implies y < b^{w-(1/n)}$$

Let A be the set of all w such that $b^w < y$, and show that $x = \sup(A)$ satisfies $b^x = y$.

If $b^x > y$, then $b^{x-(1/n)} > y$, which means that $x - \frac{1}{n}$ is a smaller lower bound, so b^x is not greater than y .
If $b^x < y$, then $b^{x+(1/n)} < y$, so $x + \frac{1}{n} \in A$, but $x < x + \frac{1}{n} \notin A$, showing a contradiction.

Thus $b^x = y$ and $x = \sup(A)$, proving the existence of a logarithm of y base b for any $y \in \mathbb{R}^+$, showing surjection. The uniqueness of x can be derived from the injection or also through supremums:

$$b^x = b^{x'} = y \implies x = \sup(A) \wedge x' = \sup(A)$$

Since the supremum is unique, then $x = x'$, showing injectivity, thus f is a bijection and therefore has an inverse which we will call $\log$. $\square$

The Logarithm

Corollary 5.11. $\log_b : \mathbb{R}^+ \setminus \{0\} \rightarrow \mathbb{R}$ such that $\log_b(y) = x$

Then the composition of $\log_b(b^x) = x$ and $b^{\log_b(x)} = x$. Here we see the $\log_b$ acting as a left and right inverse, which is the expected behavior of the inverse of a bijective function. (Remember a surjection only has a right inverse, and an injection only has a left inverse).

6 Countability and Cardinality

The natural numbers or $\mathbb{N}$ are the backbone of the idea of "counting". When we say 0, 1, 2, 3... so on and so forth, we are iterating through the naturals. We define the following:

First n naturals

Definition 6.1. The first n naturals are $[0, n) := \{k \in \mathbb{N} : k < n\}$

We can now introduce the notion of a finite set.

Finite Set

Definition 6.2. A finite set A is a set such that $\exists f, \exists n \in \mathbb{N} : f : [0, n) \rightarrow A$ and f is a bijection

An infinite set is one that is not finite, or one for which we cannot find a bijection between an interval of the first n naturals for some $n \in \mathbb{N}$.

We also introduce the notion of Dedekind infinite. A set that is Dedekind infinite is one where we can find an injective but not surjective map between the set and itself. For example:

$$S : \mathbb{N} \rightarrow \mathbb{N} : 0 \notin \text{Ran}(S) \implies S(\mathbb{N}) \subsetneq \mathbb{N}$$

So the naturals are a Dedekind infinite set under the successor operation. We can also come up with other injections such as $f(n) = n^2$. We say the following:

Dedekind Infinite

Lemma 6.3. Given a set A : A is Dedekind infinite $\implies A$ is infinite

The converse requires invoking axiom of choice as we need to assume the existence of a choice function that can generate a sequence of values to which we can apply an injective function.

If we can find a bijection f between an interval $[0, n)$ and a finite set A , then the value $n \in \mathbb{N}$ is called the cardinality of A , denoted by $|A|$. We see some inequalities related to cardinality:

Finite Sets and Cardinality

Lemma 6.4. For finite sets A and B :

1. $B \subseteq A \implies |B| \leq |A|$
2. $|A \cup B| \leq |A| + |B|$
3. $|A \times B| = |A| \cdot |B|$

We prove Lemma 6.2.1:

Proof. We define $|A|$ as $n \in \mathbb{N}, \exists f : f : [0, n) \rightarrow A \wedge \text{Im}(f) = A$. The proof is as follows:

1. $B \subseteq A \iff \forall x \in B : x \in A$ (Subset Definition)
2. $\forall x \in B : (x \in A \implies x \in \text{Im}(f))$ (By definition of f)
3. $\implies \forall x \in B \wedge A, \exists m \in [0, n) : f(m) = x$ (Definition of Preimage)
4. Thus $f^{-1}(B) = \{m \in [0, n) : f(m) \in B\}$
5. Since $\forall m \in f^{-1}(B) : m \in [0, n) \implies f^{-1}(B) \subseteq [0, n)$ (Definition of Subset)

6. Therefore either $f^{-1}(B) = [0, n) = \text{Dom}(f)$ or $f^{-1}(B) \subset [0, n)$ so it does not contain all first n naturals
7. Since f is a bijection, then $f|_{f^{-1}(B)} : f^{-1}(B) \rightarrow B$ is a bijection, so by line 6: $|B| \leq |A|$

□

We prove Lemma 6.2.2:

Proof. The proof is as follows. Since A, B are finite, axiom of choice is not required:

1. $A \cup B \iff \forall x \in A \cup B : x \in A \vee x \in B$ (Definition of Union)
2. $f : [0, n) \rightarrow A \wedge g : [0, m) \rightarrow B$ such that f, g are bijections. Then $|A| = n, |B| = m$.
3. We define a function $p : [0, m+n) \rightarrow A \cup B$ such that $\text{Ran}(p|_{[0, n)}) = A \wedge \text{Ran}(p|_{[n, m+n)}) = B$
4. If $A \cap B \neq \emptyset$, then $\exists x : x \in A \wedge x \in B$
5. $\implies x \in \text{Im}(f) \wedge \text{Im}(g)$
6. $\implies \exists l \in [0, n) \wedge \exists k \in [m, m+n) : p|_{[0, n)}(l) = p|_{[n, m+n)}(k) = x$, which means that p is not injective, so $|A \cup B| < m+n = |A| + |B|$
7. If $A \cap B = \emptyset$, then p is a bijection, so $|A \cup B| = m+n = |A| + |B|$
8. Thus $|A \cup B| \leq |A| + |B|$

□

We prove Lemma 6.2.3:

Proof. The proof is as follows:

1. We define the cartesian product of A, B as $A \times B := \bigcup \{f : [0, 2) \rightarrow A \cup B : (f(0) \in A \wedge f(1) \in B)\}$
 - (a) Remember that a function is fundamentally its graph, so each function is bijective to the set of pairs with the first element being from A and the second being from B .
 - (b) Let's restrict each f to a single, unique ordered pair so that all the graphs are disjoint.
2. Thus $|A \times B| = |\bigcup \{f : [0, 2) \rightarrow A \cup B : (f(0) \in A \wedge f(1) \in B)\}| = \# \text{ of } f$
3. Let $|A| = m \wedge |B| = n$
4. Fix an $a \in A : f(0) = a \wedge f(1) \in B \implies |\{f : [0, 2) \rightarrow A \cup B : f(0) = a \wedge f(1) \in B\}| = n$ as there are n elements in B
5. Since there are m elements in A , we have m sets $\{f : [0, 2) \rightarrow A \cup B : f(0) = a \wedge f(1) \in B\}$.
6. Thus in total we have m sets of n functions, so we have $m \cdot n$ functions in total, thus:
7. $|A \times B| = |A| \cdot |B|$

□

We note the relation of Axiom of Choice and Cartesian Products/Powers. Take for example the cartesian power $\mathbb{R}^{\mathbb{N}}$ which is the set of real valued sequences. The question is the same as the problem proposed in Lemma 6.1. To construct the sequence, we encounter the following collection:

1. $x_0 \in \mathbb{R}$
2. $x_1 \in \mathbb{R} \setminus \{x_0\}$
3. $x_2 \in \mathbb{R} \setminus \{x_0, x_1\}$

4. $x_3 \in \mathbb{R} \setminus \{x_0, x_1, x_2\}$

5. The sequences continues countably infinitely (for as many naturals there are)

Here we see that we are required to select real numbers from an infinite family of infinite sets. We encounter the same problem for an infinite family of finite sets.

We now consider the cartesian power $\mathbb{N}^{\mathbb{N}}$ which is the set of all natural-valued sequences. We see the following family:

1. $x_0 \in \mathbb{N}$

2. $x_1 \in \mathbb{N} \setminus \{x_0\}$

3. The sequence continues for all of $\mathbb{N}$

This case does not require axiom of choice however, due to the well ordering principle. We know that every set of naturals admits a minimum (an infimum that exists in the set). So we know that there exists one such possible sequence which is the naturals itself. We can also take different collections of subsets of $\mathbb{N}$, combined with the well-ordering, allowing us to define different sequences and their rules.

Question to Think About: What about $\mathbb{Q}^{\mathbb{N}}$, does such construction require Axiom of Choice?

Countable Set

Definition 6.5. A countable set A is one such that $\exists f$ such that $f : \mathbb{N} \rightarrow A$ is a bijection

We now examine countable sets. We will show results such as $\mathbb{Q}$ being countable, or the set of algebraic reals being countable, which on surface seem counterintuitive.

Theorem

Theorem 6.6. If A is countable, then $\forall B \subseteq A : B \text{ infinite} \implies B \text{ countable}$

Proof. We first see that A is infinite as well as it has an infinite subset (How can we prove this?)

Since A is countable, then we can find some bijection $f : \mathbb{N} \rightarrow A$. We then define $f^{-1}(B) = \tilde{B} \subseteq \mathbb{N}$. If B is to be countable, then $\tilde{B}$ should also be countable. We prove by induction:

Let B_0 be our base case such that $B_0 = \tilde{B}$. We know that if B is infinite then $\tilde{B}$ must be infinite (why?). We then define the following:

$$B_{k+1} = \begin{cases} \emptyset : & B_k = \emptyset \\ B_k \setminus \{\inf(B_k)\} : & B_k \neq \emptyset \end{cases}$$

Since we are dealing with sets of naturals, we do not require Axiom of Choice for this infinite family. By the well ordering principle, we can pick out the infimum of each set.

If B_{k+1} is infinite, then we see that $B_k = B_{k+1} \cup \{\inf(B_k)\}$, so B_k is also infinite. What we have done is created the countable sequence of sets:

$$\{B_k\}_{k \in \mathbb{N}} := (B_0, B_1, B_2, \dots, B_k, B_{k+1}, \dots) \text{ such that } B_k = B_{k+1} \cup \inf(B_k)$$

If we manipulate the defining proposition of the sequence, we see that $B_k \setminus B_{k+1} = \inf(B_k)$.

So if we define some function $h : \mathbb{N} \rightarrow \{B_k\}$, we get a bijection as there is a countable number of sets. We define the inf function:

$$\text{inf} : \{B_k\} \rightarrow \tilde{B}$$

We first see that inf is injective. So $\forall B_x, B_y \in \{B_n\} : \text{inf}(B_x) = \text{inf}(B_y) \implies B_x = B_y$. Imagine $B_x \neq B_y$, then $B_x \subset B_y \vee B_y \subset B_x$. Assume without loss of generality that $B_x \subset B_y$:

$$\exists y \in B_y : y \notin B_x$$

Since each B_k is infinite (implying the sets are not bounded above) and $B_x \subset B_y$, then $\forall x \in B_x : y < x$ by the ordering of the naturals. Thus y is a lower bound for B_x but $\text{inf}(B_x) \in B_x$, so $y \neq \text{inf}(B_x)$. Then:

$$y \in B_y \wedge y \neq \text{inf}(B_x) \implies \text{inf}(B_y) \leq y < \text{inf}(B_x) \quad (3)$$

The inequality $y < \text{inf}(B_x)$ follows from the fact that $y \neq \text{inf}(B_x)$ and if $y > \text{inf}(B_x)$, then $y \in B_x$ which it is not by definition. Thus $\text{inf}(B_y) < \text{inf}(B_x)$ proving the statement by contrapositive.

We now see that inf is surjective. $\forall b \in \tilde{B}, \exists B_k : \text{inf}(B_k) = b$. This follows from the well ordering property, that every subset of the naturals has an infimum. So if we pick any element of B which is a subset of the naturals, then through the recursive process, we can find a set:

$$\exists n \in \mathbb{N} : B_n := \{x \in \tilde{B} : b \leq x\}$$

The infimum of this set is b and B_n is a valid element in $\{B_k\}$ as it is either $\tilde{B} = B_0$ itself, or is a proper subset, which in that case $\exists B_m : B_m \subseteq B_0 \wedge B_n = B_m \setminus \text{inf}(B_m)$.

Now that we know inf is also a bijection, we can construct a bijection $\text{inf} \circ h : \mathbb{N} \rightarrow \tilde{B}$ which means that $\tilde{B}$ is countable. Now:

$$\exists g : g : \tilde{B} \rightarrow B \text{ such that } g \text{ is a bijection}$$

Thus $g \circ \text{inf} \circ h : \mathbb{N} \rightarrow B$, and the composition of two bijections is a bijection, showing that B is countable. $\square$

Corollary

Corollary 6.7. For any set A : A finite or countable $\iff \exists f : A \rightarrow \mathbb{N}$ such that f is an injection

Proof. We examine the forward direction first. If A is finite, then by definition 6.2 it is bijective to the subset of the first n naturals, thus f is an injection. If it is countable and not finite, then by definition 6.3, f is a bijection so it is an injection.

For the converse, if f is an injection, that means $\text{Ran}(f) \subseteq \mathbb{N}$. If $\text{Ran}(f) = \mathbb{N}$, then f is also a surjection, so A is infinite and countable.

If $\text{Ran}(f) \subset \mathbb{N}$, then if the range is infinite by Thm 6.1, it is countable, so there exists a bijection between A and $\mathbb{N}$, so A is countable.

Finally, if $\text{Ran}(f)$ is finite, then there exists a bijection between A and the first n naturals, so A is finite. $\square$

Lemma

Lemma 6.8. If A and B are countable, then $A \times B$ is countable

Proof. We define bijections $f : \mathbb{N} \rightarrow A \wedge g : \mathbb{N} \rightarrow B$. We can define a function $h : A \times B \rightarrow \mathbb{N} \times \mathbb{N}$ such that $h(a, b) = (f^{-1}(a), g^{-1}(b))$.

We see that h is a surjection: This is supported by the bijections f, g .

$$\forall (m, n) \in \mathbb{N} \times \mathbb{N}, \exists (f(m), g(n)) \in A \times B : h((f(m), g(n))) = (m, n)$$

We now prove that h is an injection: $\forall (a, b), (a', b') \in A \times B : h(a, b) = h(a', b') \implies (a, b) = (a', b')$:

$$h(a, b) = h(a', b') \implies f^{-1}(a) = f^{-1}(a') \wedge g^{-1}(b) = g^{-1}(b')$$

Since f, g are bijections, their inverses are bijections, so they are also injections, thus $a = a' \wedge b = b'$.

We now define a map $\psi : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ such that $\psi(m, n) = \frac{1}{2}(m + n)(m + n + 1) + m$. This is called the Cantor Pairing Function. We see that every natural number can be represented by this (why?).

We now prove it is an injection. Given two ordered pairs $(m, n), (m', n')$, if $m < m'$, then $\psi(m, n) < \psi(m', n')$. If $m = m' \wedge n < n'$, then we see again $\psi(m, n) < \psi(m', n')$. So ψ is strictly monotonic, therefore it is an injection as we have a total lexicographic ordering of the ordered pairs. □

Theorem

Theorem 6.9. The set $\mathbb{Q}$ is infinite and countable

Proof. If $\mathbb{Q}$ were not infinite, then $\exists t \in \mathbb{Q}, \forall r \in \mathbb{Q} : r \leq t$. Every rational can be written as $\frac{p}{q} : p, q \in \mathbb{Z} \wedge q \neq 0$. The naturals are infinite, implying the integers are infinite, so:

$$\exists p, q \in \mathbb{Z} : t = \frac{p}{q} \wedge q > 0$$

However, since $\mathbb{Z}$ is infinite, $p + 1 > p$. So we can construct a rational $s \in \mathbb{Q}$, such that:

$$s = \frac{p+1}{q} = \frac{p}{q} + \frac{1}{q} = t + \frac{1}{q}$$

Since $q > 0$, then by field properties $q^{-1} = 1/q > 0$. If $q < 0$, then we simply use $p - 1$. So we can show there exists a larger rational, thus $\mathbb{Q}$ is infinite.

We can say that the set of positive rationals is countable through the Cantor "Snake" argument by arranging them in a table of naturals minus 0 as rows and as columns. A similar thing can be done with negative integers for the negative rationals. We let ψ be the function that takes in a positive or negative rational and outputs its "place".

Each positive and negative rational will appear in the same place by construction (it has the same natural number assigned to it). We formalize it as below:

$$\forall r \in \mathbb{Q}^+ : \phi(r) = 2\psi(r) = 2n$$

Here n is the place of the positive rational in the Cantor Snake argument. For negative rationals:

$$\forall q \in \mathbb{Q}^- : \phi(q) = 2\psi(q) - 1 = 2n - 1$$

Finally $\phi(0) = 0$. So all the negative rationals map to the odd naturals, while the positive rationals map to the even naturals. We see that this is an injection as if two rationals have the same value of ϕ , then they must be both greater/less than 0.

Since the set of positive/negative rationals are bijective to the naturals, then the two rationals must be unique. Thus $\text{Ran}(\phi) \subseteq \mathbb{N}$, and because the rationals are infinite, the range is infinite. Thus by Thm 6.1, the range is countable. □

Lemma

Lemma 6.10. The set of algebraic reals: $\{x \in \mathbb{R} : (\exists n \in \mathbb{N}, \exists a_0, \dots, a_n \in \mathbb{Z} : a_n \neq 0 \wedge a_n x^n + \dots + a_1 x + a_0 = 0)\}$ is countable.

We first prove that an n degree polynomial has at most n roots:

Proof. Our base case is when $n = 0$. This polynomial does not have roots trivially.

We assume that a polynomial $p(x)$ that is degree n with at most n roots.

Then if there is a $q(x) = (x - r)p(x)$ such that $\deg(q) = \deg(p) + 1 = n + 1$, if $p(r) \neq 0$, then $\# \text{ roots}(q) = \# \text{ roots}(p) \leq \deg(p) < \deg(q)$.

If $p(r) = 0$, then $\# \text{ roots}(q) = \# \text{ roots}(p) + 1 \leq \deg(p) + 1 = \deg(q)$. Therefore for any polynomial $\# \text{ roots}(q) \leq \deg(q)$.

We can define each n degree polynomial by the $n + 1$ -tuple of its coefficients. Thus, we can define the set of polynomials degree n :

$$P_n := \mathbb{Z}^n$$

By Thm 6.1, $\mathbb{Z} \subset \mathbb{Q} \implies \mathbb{Z}$ countable. Thus the cartesian product $\mathbb{Z}^n$ is also countable as n is finite. (If the iterating set were infinite, then we would face an infinite cartesian product).

Since each polynomial $p \in P$ has at most $\deg(p)$ roots, then the set of roots for each P_n is also countable. If we define the set of roots for P_n to be T_n , then the set of all roots $\bigcup T_n$ is also countable.

Thus $\mathbb{R}_A := \bigcup T_n$, so the set of all algebraic roots is countable. □

Theorem

Theorem 6.11. There exists non-algebraic reals.

Since $\mathbb{R}_A$ is countable and $\mathbb{R}$ is not countable, then $\mathbb{R}_A \prec \mathbb{R}$, so we can define the nonalgebraic reals as:

$$\mathbb{R}_N := \mathbb{R} \setminus \mathbb{R}_A$$

We know this set is nonempty by the strict inequality for numerosity.

7 Uncountable Sets

Theorem

Theorem 7.1. The set of sequences $\{0, 1\}^{\mathbb{N}}$ is uncountable

We utilize Cantor's Diagonal argument for this theorem.

Proof. We let $f : \mathbb{N} \rightarrow \{0, 1\}^{\mathbb{N}}$. We construct a sequence $\{x_n\}_{n \in \mathbb{N}} : x_n = 1 - f(n)_n$.

Since every n th term in this new sequence is different from the n th sequence, the sequence does not belong in $\text{Im}(f)$, so f is not surjective, meaning $\{0, 1\}^{\mathbb{N}}$ is uncountable. $\square$

Lemma

Lemma 7.2. The set of functions $\{0, 1\}^{\mathbb{R}}$ is uncountable

We once again utilize Cantor's Diagonal argument for this lemma.

Proof. This time we deal with graphs of functions and not sequences. Each function $f : \mathbb{R} \rightarrow \{0, 1\}$. We also have $\phi : \mathbb{N} \rightarrow \{f\}$.

We define $h : \mathbb{R} \rightarrow \{0, 1\}$ as the function where the value of n of n th function is $(n, 1 - f_n(n))$.

We see that h does not fit in any of the functions, so the set of such functions are uncountable. $\square$

Lemma

Lemma 7.3. $\mathbb{N}^{\mathbb{N}}$ is uncountable.

Proof. Assume there to be a countable amount of natural valued sequences. We can construct a new sequence by taking the n th term of the n th sequence and adding 1 to it. Thus we come up with a new sequence not the same as the others, showing it is not countable. $\square$

Definition

Definition 7.4. Equinumerosity of two sets A, B : $A \simeq B \implies \exists f : (f : A \rightarrow B) \wedge f \text{ bijection}$

The Equinumerosity class of a set $[A]$ is the set of all sets equinumerous to A .

Theorem

Theorem 7.5. For all A , $A \notin [\mathcal{P}(A)]$

Proof. We want to prove that there exists no bijection $f : A \rightarrow \mathcal{P}(A)$. Since $\forall x \in A, \{x\} \subset A$, then $\forall x \in A : x \in \mathcal{P}(A)$ by definition of powerset. If two such singletons are equal, then by Axiom of Extensionality, their elements must be equal. Thus f is an injection.

We prove that f is not a surjection. We know that $A \subseteq A$, so $A \in \mathcal{P}(A)$, but $A \notin A$, so f is not a surjection.

If $A \in A$, then we would have a loop of infinite containment, which is not allowed by Axiom of Regularity. $\square$

Definition

Definition 7.6. We define nonequinerosity: $A \lesssim B \implies \exists f : (f : A \rightarrow B) \wedge f \text{ injection}$

Cardinality is limited to countable or finite sets, so this inequality allows us to compare numerosity or size for uncountable sets as well.

The reason why this is so is because $f(A) \simeq A$ if f is an injection (as it is one to one), and since $f(A) \subseteq B$, then $A \simeq f(A) \lesssim B$.

We observe that $A \subseteq B \implies A \lesssim B$. However $A \lesssim B \wedge B \lesssim A$ does not imply $A = B$. For example:

$$\mathbb{Q} \lesssim \mathbb{N} \wedge \mathbb{N} \lesssim \mathbb{Q}$$

Yet, $\mathbb{Q} \neq \mathbb{N}$. While we cannot make a conclusion about equality, the Cantor=Bernstein Theorem allows us to say the following:

Theorem

Theorem 7.7. $A \lesssim B \wedge B \lesssim A \implies A \simeq B$

Proof. Proof is left as an exercise to the reader. (I will complete it later, I am the reader) $\square$

However, a powerful result of this theorem is that we only need to show two injections exist between both sets to show they are equinumerous.

We build an algebraic structure out of numerosity:

Theorem

Theorem 7.8. Numerosity $\lesssim$ is a partial ordering for a set U of possible sets.

1. Reflexivity: $A \lesssim A$ is trivial as $A = A$ so $A \simeq A$
2. Anti-symmetry: $A \lesssim B \wedge B \lesssim A \implies A \simeq B$ (given by Cantor Bernstein)
3. Transitivity: $A \lesssim B \wedge B \lesssim C \implies A \lesssim C$

We prove transitivity, for which we use the following lemma:

Lemma

Lemma 7.9. Given $f : A \rightarrow B \wedge g : B \rightarrow C$. If f, g are injections, then $g \circ f$ is an injection.

Proof. By the definition of injection, we say that $\forall a, b \in B : g(a) = g(b) \implies a = b$. By preimage we say that:

$$\exists x, y \in A : f(x) = a \wedge f(y) = b$$

So our initial proposition becomes $\forall x, y \in A : g(f(x)) = g(f(y)) \implies f(x) = f(y)$. By injectivity of g , we get that $a = b$, so $g \circ f$ is an injection. $\square$

Now the fact that $g \circ f$ is an injection tells us by the definition of numerosity (nonequinerosity), that $A \lesssim C$, proving transitivity.

We now prove the following lemma about the Cartesian product (which will contribute as the operation to our algebraic structure):

Lemma

Lemma 7.10. $\forall A, B, C \in U : A \lesssim B \implies A \times C \lesssim B \times C$

We are given $A \lesssim B$ which means we have an injection $f : A \rightarrow B$. We can define the following function:

$$\forall (a, c) \in A \times C : g(a, c) = (f(a), c)$$

We see that this is an injection. Take two ordered pairs $(a, c), (b, d)$, then:

$$g(a, c) = g(b, d) \implies (f(a), c) = (f(b), d) \implies f(a) = f(b) \wedge c = d$$

We derive the equivalence of the components by our construction of ordered pairs with Kuratowski pairs. It also stems from the fact that the two components are from different sets. By the injectivity of f :

$$f(a) = f(b) \wedge c = d \implies a = b \wedge c = d \implies (a, c) = (b, d)$$

Thus g is injective so we see that $A \times C \lesssim B \times C$. The same proof can be carried out to show that $C \times A \lesssim C \times B$. We will see that the inequality does not behave nicely with $\lesssim$.

This likely seems similar to the ordered preserving operations on ordered fields, hinting at an underlying algebraic structure.

Definition

Definition 7.11. A partially ordered semigroup is a group equipped with associativity, identity and order ($\leq$), but not with inverses under the group operation.

Additionally $\forall a, b, c \in G : a \leq b \implies a + c \leq b + c \iff c + a \leq c + b$

We see that the cross product $\times$ (our group operation) is associative (we can find a bijection between $A \times (B \times C)$ and $(A \times B) \times C$).

By Lemma 7.3, we see that the order is preserved under the operation. We do not have inverse for the cartesian product. For the identity, let's call it I :

$$A \times I \simeq A$$

This happens when I has cardinality of 1, which means we can have multiple identities. However, the identity of a group must be unique! So we are not dealing with sets but rather equivalence classes of equinumerous sets!

The identity I is the class of all sets cardinality 1. Then we get the following property: $[A] \times [B] \simeq [A \times B]$. So when we say that $A \lesssim B$, what we really mean is $[A] \lesssim [B]$, as:

$$A' \in [A] \wedge B' \in [B] \wedge \exists f : A \rightarrow B \wedge f \text{ injection}$$

Then $A' \in [A] \wedge B' \in [B]$ by Definition 7.1 means we can find bijections g, h between A', A and B', B respectively. Then, we get the following injection:

$$g \circ f \circ h : A' \rightarrow B' \implies A' \lesssim B'$$

Thus the choice of representative is arbitrary, but when we say a numerosity inequality is true for certain sets, it is also true for the sets equinumerous to them.

Definition

Definition 7.12. The equinumerosity class $[\mathbb{R}]$ is called the continuum.

We can show that $\mathcal{P}(\mathbb{N}) \in [\mathbb{R}]$.

Proof. $\mathcal{P}(\mathbb{N}) \simeq \{0, 1\}^{\mathbb{N}}$ via the following bijection:

$$\forall A \in \mathcal{P}(\mathbb{N}) : f_n(A) := \begin{cases} 0 & n \notin A \\ 1 & n \in A \end{cases}$$

If two sequences are equal, then we know that their preimages must be the same, so f is injective. We also see f is surjective, since every sequence is the size of $\mathbb{N}$, it obtains an equivalent expression as a subset of $\mathbb{N}$.

Before we derived an injection from $\{0, 1\}^{\mathbb{N}} \rightarrow [0, 1]$, thus we can derive an injection from $\mathcal{P}(\mathbb{N}) \rightarrow [0, 1]$. Since $[0, 1] \subset \mathbb{R}$, the injection follows, so $\mathcal{P}(\mathbb{N}) \lesssim \mathbb{R}$.

Now $\mathbb{R}$ is the set of Dedekind cuts to $\mathbb{R} \lesssim \mathcal{P}(\mathbb{Q})$.

Since $\mathbb{Q} \simeq \mathbb{N}$ then $\mathcal{P}(\mathbb{Q}) = \mathcal{P}(\mathbb{N})$ (why? Technically by 2^n rule but not proven). Thus $\mathbb{R} \lesssim \mathcal{P}(\mathbb{N})$, so:

$$\mathcal{P}(\mathbb{N}) \simeq \{0, 1\}^{\mathbb{N}} \simeq [0, 1] \simeq \mathbb{R}$$

□

Lemma

Lemma 7.13. $\mathbb{R} \times \mathbb{R} \simeq \mathbb{R}$ and $\forall n \in \mathbb{N} : \mathbb{R}^n \simeq \mathbb{R}$

We first prove that $\mathcal{P}(\mathbb{N}) \times \mathcal{P}(\mathbb{N}) \simeq \mathcal{P}(\mathbb{N})$. We see trivially that $\mathcal{P}(\mathbb{N}) \lesssim \mathcal{P}(\mathbb{N}) \times \mathcal{P}(\mathbb{N})$.

$$\mathcal{P}(\mathbb{N}) \times \mathcal{P}(\mathbb{N}) \simeq \{0, 1\}^{\mathbb{N}} \times \{0, 1\}^{\mathbb{N}}$$

We can construct an injection $f : \{0, 1\}^{\mathbb{N}} \times \{0, 1\}^{\mathbb{N}} \rightarrow \{0, 1\}^{\mathbb{N}}$. If we take a pair of sequences, simply making the elements from the first sequence as odd elements and from the second as even elements, we get a unique sequence. Therefore:

$$\mathcal{P}(\mathbb{N}) \times \mathcal{P}(\mathbb{N}) \simeq \{0, 1\}^{\mathbb{N}} \times \{0, 1\}^{\mathbb{N}} \lesssim \{0, 1\}^{\mathbb{N}} \simeq \mathcal{P}(\mathbb{N})$$

Thus $\mathcal{P}(\mathbb{N}) \times \mathcal{P}(\mathbb{N}) \simeq \mathcal{P}(\mathbb{N})$.

Proof. We note that $\mathbb{R} \lesssim \mathbb{R} \times \mathbb{R}$ trivially, as the real numbers are embedded in a 2D Euclidean space.

$$\mathbb{R} \times \mathbb{R} \lesssim \mathcal{P}(\mathbb{Q}) \times \mathcal{P}(\mathbb{Q}) \simeq \mathcal{P}(\mathbb{N}) \times \mathcal{P}(\mathbb{N}) \simeq \mathcal{P}(\mathbb{N}) \simeq \mathbb{R}$$

Thus $\mathbb{R} \times \mathbb{R} \lesssim \mathbb{R}$, so $\mathbb{R} \times \mathbb{R} \simeq \mathbb{R}$.

□

Proof. We now prove that $\forall n \geq 1 : \mathbb{R}^n \simeq \mathbb{R}$. The inequality $\mathbb{R} \lesssim \mathbb{R}^n$ is trivial. We prove the other direction:

$$\mathbb{R}^n \lesssim \mathcal{P}(\mathbb{Q})^n \simeq \mathcal{P}(\mathbb{N})^n$$

Since $n \in \mathbb{N}$, if n is even, then we create $n/2$ number of pairs of $\mathcal{P}(\mathbb{N})$. Using the property initially proved, we get:

$$\mathcal{P}(\mathbb{N})^n \simeq \mathcal{P}(\mathbb{N})^{n/2} \simeq \mathcal{P}(\mathbb{N})^{n/4} \simeq \dots$$

We repeat this process till we get $\mathcal{P}(\mathbb{N})$. If n is odd, then we apply this process to $n - 1$. We then get:

$$\mathcal{P}(\mathbb{N})^n \simeq \mathcal{P}(\mathbb{N})^{n-1} \times \mathcal{P}(\mathbb{N}) \simeq \mathcal{P}(\mathbb{N}) \times \mathcal{P}(\mathbb{N}) \simeq \mathcal{P}(\mathbb{N})$$

Therefore we complete our previous inequality:

$$\mathbb{R}^n \lesssim \mathcal{P}(\mathbb{Q})^n \simeq \mathcal{P}(\mathbb{N})^n \simeq \mathcal{P}(\mathbb{N}) \simeq \mathbb{R}$$

Now that we have both inequalities, we have proven that $\mathbb{R}^n \simeq \mathbb{R}$.

□

Lemma

Lemma 7.14. $\mathbb{R}^{\mathbb{N}} \simeq \mathbb{R}$

We know that $\mathbb{R} \simeq \{0, 1\}^{\mathbb{N}}$, so by Lemma 6.3 and 7.2:

$$\mathbb{R}^{\mathbb{N}} \simeq \{0, 1\}^{\mathbb{N} \times \mathbb{N}} \simeq \{0, 1\}^{\mathbb{N}} \simeq \mathcal{P}(\mathbb{N}) \simeq \mathbb{R}$$

Lemma

Lemma 7.15. $\mathbb{R} \lesssim \mathbb{R}^{\mathbb{R}}$

We first prove that $\mathbb{N} \times \mathbb{R} \simeq \mathbb{R}$. The inequality $\mathbb{R} \lesssim \mathbb{N} \times \mathbb{R}$ is trivial. We prove the other direction by Lemma 7.2:

$$\mathbb{N} \times \mathbb{R} \simeq \mathbb{N} \times \mathcal{P}(\mathbb{N}) \lesssim \mathcal{P}(\mathbb{N}) \times \mathcal{P}(\mathbb{N}) \simeq \mathcal{P}(\mathbb{N}) \simeq \mathbb{R}$$

Proof. We can now track the following inequality:

$$\mathbb{R} \lesssim \mathcal{P}(\mathbb{R}) \simeq \{0, 1\}^{\mathbb{R}} \simeq \{0, 1\}^{\mathbb{N} \times \mathbb{R}} \simeq \mathcal{P}(\mathbb{N})^{\mathbb{R}} \simeq \mathbb{R}^{\mathbb{R}}$$

□

8 Metric Spaces

We earlier defined a sequence as the graph of a function $f : \mathbb{N} \rightarrow A$. While this is the underlying machinery, we consider a sequence as a set of elements indexed by the naturals, denoted as $\{x_n\}_{n \in \mathbb{N}}$.

First we deal with "real-valued sequences", which means that all $x_n \in \mathbb{R}$. We define two important notions, in the context of the reals:

Definition

Definition 8.1. A Cauchy sequence: $\forall k \in \mathbb{N}, \exists n_0 \in \mathbb{N}, \forall m, n \geq n_0 : |x_m - x_n| < \frac{1}{k+1}$

The reason we set the bound as $\frac{1}{k+1}$ is that the bound will always exist in the reals and as we increase k , the bound will get smaller, which is the behavior we want to model. We could set the bound to $\frac{1}{r+1}$, where r is a real, but the natural numbers allow us to create a sequence of bounds unlike the reals which are uncountable.

Definition

Definition 8.2. A convergent sequence: $\exists L \in \mathbb{R}, \forall k \in \mathbb{N}, \exists n_0 \in \mathbb{N}, \forall n \geq n_0 : |x_n - L| < \frac{1}{k+1}$

This is different from a Cauchy sequence as we are now fixing a set "limit point". As we decrease the bound, we can always find some element in the sequence arbitrarily close. So for a very large k we see that x_n tends to L .

Example: We show the following is Cauchy and convergent:

$$a_0 = 1 \wedge a_{n+1} = 3 - \frac{1}{a_n}$$

We show that $\forall n \in \mathbb{N} : 1 \leq a_{n+1} \leq 3$ and $a_n < a_{n+1}$:

$$1 \leq a_n \leq 3 \implies -1 \leq -\frac{1}{a_n} \leq -\frac{1}{3}$$

$$a_{n+1} \leq 3 - \frac{1}{a_n} \leq 3 - \frac{1}{3} < 3$$

Also $a_n \geq 1 \implies a_{n+1} > a_n \geq 1$, so $1 \leq a_{n+1} \leq 3 \wedge a_n < a_{n+1}$. Since each consecutive term is greater, we call this monotone (monotonically increasing).

We now prove the sequence is Cauchy:

Proof. We need to show we can find some n_0 such that $\forall k \in \mathbb{N}$ the proposition holds true. We apply the triangle inequality:

$$\begin{aligned} |a_{n+1} - a_n| &\leq \left| 3 - \frac{1}{a_n} - \left(3 - \frac{1}{a_{n-1}} \right) \right| = \left| \frac{1}{a_{n-1}} - \frac{1}{a_n} \right| \\ &= \left| \frac{a_n - a_{n-1}}{a_n a_{n-1}} \right| \leq \frac{1}{2} |a_n - a_{n-1}| \leq \frac{1}{2^n} |a_1 - a_0| = \frac{1}{2^n} < \frac{1}{k+1} \end{aligned}$$

We see that $2^{n_0} > k+1$. We know that $\forall n \in \mathbb{N} : n+1 \leq 2^n$, so if we set $n_0 := k+1$, then $k+1 < 2^{k+1}$ and $\frac{1}{2^{k+1}} = \frac{1}{k+1}$. $\square$

8.1 Metrics

We now generalize the notion of distance to the concept of a "metric":

Definition

Definition 8.3. A metric space (X, ϱ) is a set X equipped with a metric $\varrho : X \times X \rightarrow \mathbb{R}^+$:

1. $\forall x, y \in X : \varrho(x, y) \geq 0 \wedge \varrho(x, x) = 0$ (Positivity)
2. $\forall x, y \in X : \varrho(x, y) = \varrho(y, x)$ (Symmetry)
3. $\forall x, y, z \in X : \varrho(x, z) \leq \varrho(x, y) + \varrho(y, z)$ (Triangle Inequality)

We will now look at various metrics. The typical Euclidean metric (distance between two points in $\mathbb{R}^n$) is defined as:

$$\left(\sum_{i=1}^n |x_i - y_i|^2 \right)^{\frac{1}{2}}$$

We can in fact generalize this notion to create p -metrics for $\uparrow^p$ spaces (which are spaces of X valued sequences with a p -norm. Think, vector spaces with a fixed basis such that the elements can be represented as finite sequences, or infinite with Axiom of Choice):

$$p\text{-metric} := \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{\frac{1}{p}}$$

The ∞ -metric is then defined as the following and is referred to as the "box metric":

$$\max_{i=1,2,\dots,n} |x_i - y_i|$$

We address the "shapes" of these metrics. This construct stems from the idea of a norm, attached to a vector space (which creates a normed vector space):

Definition

Definition 8.4. A norm $\|\cdot\|$ is a metric that gives the magnitude of a vector from the "origin" or 0 element. Let V be a vector space:

1. $\forall v \in V : \|v\| \geq 0$ (Positivity)
2. $\forall v \in V \wedge c \in F : \|cv\| = |c| \|v\|$ (Scalar Multiplication)
3. $\forall v, w \in V : \|v + w\| \leq \|v\| + \|w\|$ (Triangle Inequality)

The norm pretty closely mirrors the properties of a metric space (that is because it is, or it induces one). We can derive the remaining properties of the metric space (which is symmetry) by the following:

$$\forall v, w \in V : \|v - w\| = \|-(w - v)\| = |-1| \|w - v\| = \|w - v\|$$

This completes all the properties needed for the vector space to be a metric space. We can now draw out the p -norm unit circles (which are the set of points/vectors a distance 1 away from the origin): We will return to verifying the p -metrics to be valid metrics via the norms. The p -norm now becomes the distance between the origin and the vector:

$$\forall v \in V : \|v\|_p = \left(\sum_{k=1}^n (|v_k|^p) \right)^{\frac{1}{p}}$$

With the introduction of metrics, we can now generalize the notion of Cauchy and convergent sequences for an X valued sequence $\{x_n\}_{n \in \mathbb{N}}$ with a metric (X, ϱ) :

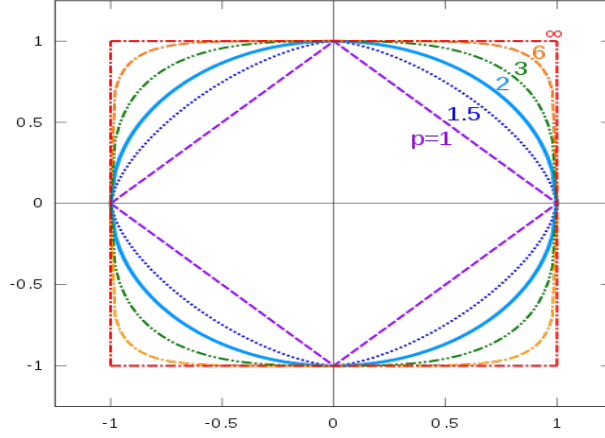


Figure 1: Unit circles of p -norms

Definition

Definition 8.5. A Cauchy sequence: $\forall \epsilon > 0, \exists n_0 \geq 0, \forall m, n \geq n_0 : \varrho(x_m, x_n) < \epsilon$

A convergent sequence: $\exists L \in X, \forall \epsilon > 0, \exists n_0 \geq 0, \forall n \geq n_0 : \varrho(x_n, L) < \epsilon$

The definition is largely similar, with the difference being an arbitrary upper bound ϵ as it is not specified what the set X is and what the metric ϱ is.

Lemma

Lemma 8.6. A convergent sequence is Cauchy.

Proof. Given (X, ϱ) and a sequence $\{x_n\}_{n \in \mathbb{N}}$, we let the limit point be L . Then:

$$\forall \epsilon > 0, \exists n_0 \geq 0, \forall n \geq n_0 : \varrho(x_n, L) < \epsilon/2$$

We choose an $m \geq n_0$ such that $\varrho(x_m, L) < \epsilon/2$. Then by the triangle inequality:

$$\varrho(x_n, x_m) \leq \varrho(x_n, L) + \varrho(L, x_m) = \varrho(x_n, L) + \varrho(x_m, L) = \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

Thus $\forall \epsilon > 0$, we have shown there exists an n_0 , which is the same as our initial one in the limit definition, such that $\forall m, n \geq n_0 : \varrho(x_m, x_n) < \epsilon$. $\square$

Lemma

Lemma 8.7. For (X, ϱ) , an X valued, convergent sequence $\{x_n\}_{n \in \mathbb{N}}$ has at most one limit L .

Proof. We assume there to be two limits L and L' such that $x_n \rightarrow L \wedge x_n \rightarrow L'$. Since $L \neq L'$, then we can find an ϵ such that $\varrho(L, L') > \epsilon$. Then we define an n_0 , such that:

$$\forall n \geq n_0 : \varrho(x_n, L) < \frac{\epsilon}{2} \wedge \varrho(x_n, L') < \frac{\epsilon}{2} \quad (4)$$

By positivity we know that $\varrho(x, x) = 0$, so by the triangle inequality:

$$\varrho(L, L') \leq \varrho(x_n, L) + \varrho(x_n, L') = \epsilon < \varrho(L, L')$$

We get that $\varrho(L, L') < \varrho(L, L')$ which is a contradiction by metric properties, so $L = L'$. $\square$

Definition

Definition 8.8. A subsequence is defined by $\{x_{n_k}\}_{n_k \in \mathbb{N}}$ where $\{n_k\}_{k \in \mathbb{N}}$ is a strictly increasing/monotonically increasing sequence

Typically we can define the n_k sequence by some rule, such as even naturals or odd naturals.

Lemma

Lemma 8.9. For a Cauchy sequence $\{x_n\}_{n \in \mathbb{N}}$, if there is a convergent subsequence such that $x_{n_k} \rightarrow z \implies x_n \rightarrow z$ where z is a limit point

Proof. The sequence being Cauchy means $\forall \epsilon > 0, \exists n_0 \geq 0, \forall m, n \geq n_0 : \varrho(x_m, x_n) < \epsilon/2$.

For the convergent subsequence, we get that $\exists z \in X, \forall \epsilon > 0, \exists n_1 \geq 0, \forall n \geq n_1 : \varrho(x_n, z) < \epsilon/2$.

Then we choose an $m \geq n_1 \wedge m \geq n_0 \wedge n \geq n_0 \wedge n \geq n_1$. By triangle inequality:

$$\varrho(x_m, z) \leq \varrho(x_m, x_n) + \varrho(x_n, z) = \epsilon$$

Thus the sequence for any $m \geq n_1 \wedge m \geq n_0$ converges to z . We can simply set the $n_2 = \max\{n_0, n_1\}$ such that:

$$\exists z \in X, \forall \epsilon > 0, \exists n_2 \geq 0, \forall n \geq n_2 : \varrho(x_n, z) < \epsilon$$

□

Intuition Note: We use $\epsilon > 0$, as if $\epsilon = 0$, then that would mean the two values being compared are equal. However if you take the n th term for a very very large n in a sequence, not necessarily you will reach the limit, but you may be ever so close, which is why there will be some small difference ϵ . However, if we say it equals ϵ and then choose the next term of the sequence, then the difference will not necessarily be equal to. However, if it is converging, then it will likely be less, so we say $< \epsilon$. If we choose the ϵ then we can guarantee a closer term to the limit always.

What if the sequence does actually reach the limit? We view the discrete metric for a point y :

$$\varrho(x, y) := \begin{cases} 0 & : x = y \\ 1 & : x \neq y \end{cases}$$

Corollary

Corollary 8.10. If a sequence in (X, ϱ) is Cauchy, then it is convergent and eventually constant.

Proof. If it is Cauchy, then $\forall \epsilon > 0, \exists n_0 \geq 0, \forall m, n \geq n_0 : \varrho(x_m, x_n) < \epsilon$. For any epsilon (eg $\epsilon = 1$), the metric being less than epsilon implies that $x_m = x_n$. So we can define a limit L such that $x_m = x_n = L$.

We can see that the sequence does meet and stagnates at the limit under this metric.

□

Lemma

Lemma 8.11. Urysohn's Subsequence Principle says the following are equivalent for a sequence $\{x_n\}_{n \in \mathbb{N}}$ and a limit point z :

1. $x_n \rightarrow z$
2. Every subsequence $\{x_{n_k}\}_{n_k \in \mathbb{N}}$ contains a convergent subsubsequence such that $x_{n_{k_j}} \rightarrow z$

Proof. We prove equivalence of the statements by showing an if and only if relation:

$\Rightarrow$: By lemma 8.1, we get that the sequence is Cauchy. We apply Lemma 8.3 twice to see that there exists such a subsubsequence converging to z .

$\Leftarrow$: We proceed by contrapositive. Assume $\{x_n\}_{n \in \mathbb{N}}$ does not converge at all, then $\forall \epsilon > 0, \exists n_0 \geq 0, \forall n \geq n_0 : \varrho(x_n, z) \geq \epsilon$. If we then set:

$$n_0 = 1 \wedge n_k := \inf\{m > n_k \wedge \varrho(x_m, z) > \epsilon\}$$

Essentially each n_k of the subsequence is the smallest value of m so that the sequence does not converge, then the subsequence does not converge. So by the contrapositive of lemma 8.3, the subsubsequence does not converge. $\square$

Note 8.1. Tips for Convergence Proofs: If the n term is in the exponent, eg $1/n$ try to use archimedian to bring it to the numerator. For example:

$$\begin{aligned} \forall \epsilon > 0, \exists n \in \mathbb{N} \setminus \{0\} : b^{1/n} - 1 < \epsilon \\ b^{1/n} - 1 \leq b - 1 < \epsilon &\implies \frac{\epsilon}{b-1} > 0 \implies \exists n : n \cdot \frac{\epsilon}{b-1} > 1 \\ &\implies n > \frac{b-1}{\epsilon} \implies n := \frac{b-1}{\epsilon} + 1 \end{aligned}$$

Also if we are trying to show less than or equal to inequalities, we want to find a **LARGER** numerator OR a **SMALLER** denominator.

To determine monotonicity of a sequence (or subsequence) with a recursive definition:

1. Check the base case e.g. $a_0 < a_1$
2. Assume P_n such that $a_{n-1} < a_n$
3. Write the (recursive) sequence as a function and show it is increasing or decreasing
4. Apply it to P_n to show $a_n < a_{n+1}$ is true, thus concluding for all n
5. This can help us eliminate certain candidates for limits

Midterm 2 Problem 5: Prove that $d(x, y) := \inf\{\varrho(x, w) + \varrho(w, y) : w \in X\}$ is a metric.

Proof. We will skip over symmetry and positivity (those are easy to prove). For the triangle inequality, we need to prove the following:

$$d(x, y) \leq d(x, z) + d(z, y) \implies \inf\{\varrho(x, w) + \varrho(w, y)\} \leq \inf\{\varrho(x, w) + \varrho(w, z)\} + \inf\{\varrho(z, w) + \varrho(w, y)\}$$

We know that by infimum properties that $\inf(A) + \inf(B) = \inf(A + B)$. So:

$$\inf\{\varrho(x, w) + \varrho(w, z)\} + \inf\{\varrho(z, w) + \varrho(w, y)\} = \inf\{\varrho(x, w) + \varrho(w, y) + 2\varrho(w, z)\}$$

We need to prove the infimum of the left set is less than the infimum of the right hand set. We see that by positivity $\varrho(w, z) \geq 0$:

$$\forall w \in X : \varrho(x, w) + \varrho(w, y) \leq \varrho(x, w) + \varrho(w, y) + 2\varrho(w, z)$$

The left hand side of the inequality is therefore always a bound for the elements of the right hand set so:

$$\forall w \in X : \inf\{\varrho(x, w) + \varrho(w, y)\} \leq \varrho(x, w) + \varrho(w, y) \leq \inf\{\varrho(x, w) + \varrho(w, y) + 2\varrho(w, z)\}$$

Thereby proving the triangle inequality for this metric. □

9 Topology

9.1 Open and Closed sets (In Metric Spaces)

We now deal with the notion of a set being "open" and "closed". This is the same as saying does the set contain its boundary points or not (however we have not defined boundary points).

Definition

Definition 9.1. An open ball around point x , radius r in (X, ϱ) is $B(x, r) := \{y \in X : \varrho(x, y) < r\}$

We talked about convergent and Cauchy sequences using the idea of a metric (distance between points) given to a set X .

With the Euclidean metric p_2 , the ball in 1D is an open interval. In 2D it is a circle, and in 3D a sphere, and so on so forth. We saw that in 2D for the $\|\cdot\|_\infty$, we get the shape of a square.

This is equivalent to saying the open ball is the cartesian product of two open intervals, so in $(\mathbb{R}^n, p_\infty)$:

$$B_\infty(x, r) = \bigtimes_{i=1}^n (x_i - r, x_i + r)$$

We can use this to see if a set is open:

Definition

Definition 9.2. An open set is one for which we can find an empty ball around each point that is a subset of the set

Note the IFF: A set is closed if and only if its complement is open.

Corollary

Corollary 9.3. A set A is open if and only if its complement is closed.

$\Leftarrow$: A^c being closed means its complement is open by Definition 9.2. $(A^c)^c = A$, so A is open.

$\Rightarrow$:

Lemma

Lemma 9.4. The singleton $\{x\}$ is a closed set.

Assume $x \in (X, \varrho)$, then for any $r \in \mathbb{R}$, we can find some point $y \in X$ such that $\varrho(x, y) = r$. If we then set the open ball around x to be $B(y, r)$, then it will not contain x by definition of open ball.

We can construct such open balls for all $y \in X \setminus \{x\}$ so by definition 9.2, $X \setminus \{x\}$ is open, thus $\{x\}$ is closed.

Lemma

Lemma 9.5. Every "open" ball is open

We take any point in the open ball $y \in B(x, r)$. By definition $\varrho(x, y) < r$, we can then define an open ball around y :

$$B(y, r') := \{z \in X : \varrho(y, z) < r'\}$$

Then $\varrho(y, z) < r' \wedge r' + \varrho(x, y) < r$. If we set $\varrho(y, z) < r - \varrho(x, y)$. By triangle inequality:

$$\varrho(x, z) \leq \varrho(x, y) + \varrho(y, z) < \varrho(x, y) + r - \varrho(x, y) = r$$

Since $\varrho(x, z) < r$, then by definition $z \in B(x, r)$, so we can always find some open ball around each point in the open ball such that it belongs to $B(x, r)$.

Remember that our lemma is equivalent to the statement that we can find an open ball around each point in parent ball such that it is a subset of the parent ball. So we need to find a radius of the smaller ball around each point that allows us to say that.

Lemma

Lemma 9.6. If ϱ is a discrete metric on X , then every subset of X is open and closed.

If we consider the open ball $B(x, r)$ with $0 < r < 1$, then the only point contained within this ball is the singleton $\{x\}$. By lemma 9.1, this is closed, however, since its an open ball it is open as well.

For any subset $A \subseteq X$, if we take the discrete points of A as singletons, then each point in A has an open ball that is a subset of A . Thus by Definition 9.2, A is open.

We can depend on the same construction for $X \setminus A$ which becomes open, and since $A^c = X \setminus A$, then A is a closed set by Definition 9.2.

Lemma

Lemma 9.7. A closed ball $\bar{B}(x, r) := \{y \in X : \varrho(y, x) \leq r\}$ is closed

The complement of this set is $A := \{y \in X : \varrho(y, x) > r\}$. We aim to show that this set is open:

$$\forall y \in A : B(y, r') := \{z \in X : \varrho(y, z) < r'\}$$

We define open balls in $A = \bar{B}^c$ and just like Lemma 9.2 find a suitable radius r' . By the triangle inequality:

$$\varrho(x, y) \leq \varrho(x, z) + \varrho(y, z) \implies \varrho(x, y) - \varrho(y, z) \leq \varrho(x, z)$$

Using the inequality $\varrho(y, z) < r'$, we see that:

$$r - r' < \varrho(x, y) - r' < \varrho(x, z)$$

So if we set $0 < r' < \varrho(x, y) - r$, we see that $\varrho(x, z) > r$ which means $z \in A \vee B(y, r') \subseteq A$. We also know that $\varrho(x, y) - r > 0$ by definition of A . Since A is open, then $\bar{B} = A^c$ is closed.

9.2 Topologies

Let us provide some motivation behind the notion of a topology. By Definition 9.2, a set is closed if its complement is open. How do we define an open set? Perhaps an open set is one whose complement is closed. We arrive at a circular definition.

Let's try a different approach. A closed set is one which has its boundary in it, and an open set is one whose complement is closed. How do we define the boundary? Perhaps we can define it as the set without its inside

(interior). How do we define the interior of a set?

Perhaps (and this is not formal) we can call it the largest open proper subset of the closed set. How do we define this maximal subset? It is maximal in relation to other open subsets, so we need a general definition of an open set.

Definition

Definition 9.8. A topology τ is a collection of subsets of a space X which are open if:

1. $\emptyset, X \in \tau$
2. $\bigcup A \in \tau$ (Closed under Arbitrary Unions - finite/infinite)
3. $\bigcap A \in \tau$ (Closed under **finite** intersection)

If all these requirements are satisfied, then the collection of subsets is called a topology τ on X . The "coarsest" topology is simply $\{\emptyset, X\}$, which can be checked follows all definitions.

Definition

Definition 9.9. A topology can also be defined as a collection of closed sets such that:

1. Since $X = \emptyset^c$, and since X is open, then $\emptyset$ is closed. Similarly since $\emptyset$ is open, then X is closed.
2. $(\bigcup A)^c = \bigcap A^c$. A^c is closed, since A is open, and similarly $(\bigcup A)^c$ is closed. Therefore, the arbitrary intersection of closed sets belongs in the topology.
3. Similarly $(\bigcap A)^c = \bigcup A^c$, so the finite union of closed sets belongs to the topology.

Building upon the notion of topology, we can often "define" or "generate" a topology through a smaller collection of subsets of X , which we call the basis.

Definition

Definition 9.10. A basis $\mathcal{B}$ of a topology τ is a collection of X :

1. $\forall x \in X, \exists B \in \mathcal{B} : x \in B \wedge B \subseteq X$
2. $\forall B_1, B_2 \in \mathcal{B}, \forall x \in B_1 \cap B_2, \exists B_3 \in \mathcal{B} : x \in B_3 \subseteq B_1 \cap B_2$

The first part of the definition ensures that every point in the space is "covered" by at least one basis element (if not more depending on the space). That way, we can describe a topology on X adequately (all of X must be in the topology).

We will touch on the purpose of the second definition in the following lemma:

Lemma

Lemma 9.11. A topology τ **generated** by a basis $\mathcal{B}$ is defined as:

$$\tau := \{U \subseteq X : (\forall x \in U, \exists B \in \mathcal{B} : x \in B \subseteq U)\}$$

Here we define a topology "generated" by a basis as the collection of subsets that meet the criteria of openness. While our previous criteria for openness was the topological axioms, we introduce a more simple construct of a basis to define openness.

We see that each basis element is trivially open by this definition as it is a subset of X and a subset of itself. The reader can check that this corresponds to the definitions given in the previous section of open/closedness in a metric space. So the basis for metric spaces is the collection of all open ball subsets of (X, ϱ) .

We see that such generated topology does follow the topological axioms (so if we can come up with a collection/basis that satisfies Defn 9.5, then we fulfill the topological axioms).

The following lemma will talk about the topology induced by a metric on a space specifically, but the proof can be generalized to non metrizable spaces.

Lemma

Lemma 9.12. A metric ϱ on X induces a topology $\tau := \{O \subseteq X : O \text{ open}\}$

For the first statement, we see that $\emptyset$ is trivially open as we can always define open balls that are subsets of it since it contains no elements. X is also trivially open as any open ball must be a subset of it (there exists nothing beyond X).

Given a collection of open subsets A , then:

$$x \in \bigcup A \implies \exists A_i \in A : x \in A_i \wedge A_i \text{ open}$$

So since A_i is open, we can find an open ball $B(x, r) \subseteq A_i$, so $\forall x \in \bigcup A, \exists B(x, r) \subseteq A_i \subseteq \bigcup A$. Thus the union is open.

Now given the intersection $\bigcap A$, then:

$$\forall x \in \bigcap A, \forall A_i \in A : x \in A_i$$

Since each A_i is open, then we can find an open ball $B(x, r_i)$ for each subset such that $B(x, r_i) \subseteq A_i$. We see that the balls are nested:

$$B(x, r_0) \subseteq B(x, r_1) \subseteq \dots B(x, r_i)$$

If $r_0 := \min\{r_i\}$, then $\forall A_i : B(x, r_0) \subseteq A_i$, so for any point in the intersection, we can find an open ball contained in the intersection, meaning the intersection is open.

To generalize this part, we know that $\forall x \in A_i, \exists B(x, r_i)$. Now this is where the second part of the basis definition plays a role:

$$\forall x \in A_1, A_2, \exists B_1, B_2 : x \in B_1 \subseteq A_1 \wedge x \in B_2 \subseteq A_2$$

$$\forall x \in B_1 \cap B_2, \exists B_3 : x \in B_1 \wedge x \in B_2 \implies x \in A_1 \wedge x \in A_2 \implies x \in B_3 \subseteq B_1 \cap B_2 \subseteq A_1 \cap A_2$$

So for any point in the intersection, we can find a basis element (in this case open ball) that belongs in the intersection. We repeat the above process inductively (as we only deal with finite intersections), showing us that the finite intersections are also open. e of the set relies on arbitrary intersections which is possible due to the arbitrary intersections of closed set collections in the topology.

We can also say the following about a topology generated by a basis:

Lemma

Lemma 9.13. The topology τ generated by the basis $\mathcal{B}$ is equivalent to the collection of unions of subcollections of $\mathcal{B}$.

Proof. We call the topology of unions τ_B . We assume that $\mathcal{B} \subseteq \tau_B$. Then any arbitrary unions and finite intersections of basis elements belong in the topology. By our initial definition of τ and the basis, $\mathcal{B} \subseteq \tau$ and

so its arbitrary unions and intersections also belong on the topology.

To show the two topologies are equivalent, we aim to show all open sets in τ are unions of basis elements. In essence, both the topologies are equally fine as the arbitrary unions and finite intersections of $\mathcal{B}$ would be equivalent representations of the existing sets in τ defined by the first definition.

We know by the definition of $U \in \tau$:

$$\begin{aligned} \forall x, \forall B_x \subseteq U : x \in U &\implies \exists B_x : x \in B_x \\ \forall x : x \in U &\implies x \in \bigcup B_x \iff U \subseteq \bigcup B_x \end{aligned}$$

We do not require Axiom of Choice here as the property of bases allows us to generate a (not necessarily unique) collection of basis elements.

Now we want to prove $\bigcup B_x \subseteq U$. We can either use the collection of all basis elements that are subsets of U (this does not require choice as we have a rule), so if x is in its union, then x is one of the subsets which means x is in U .

Alternatively, we can select one such collection where each $x \in U$ belongs to one specific basis element which is a subset of U . Then:

$$x \in \bigcup B_x \implies \exists B_x \in \mathcal{B} : x \in B_x \implies x \in U$$

Thus $\bigcup B_x \subseteq U$, thus $U = \bigcup B_x$, showing that the two topologies τ_B and τ are equivalent. (To summarize, we defined the first topology τ_b and then showed that any element in τ has an equivalent representation in τ_B).

The reason why the second approach required axiom of choice is because while we can guarantee a basis element which is a subset of U containing x for every $x \in U$, we cannot guarantee ONE, UNIQUE element (there can be multiple). Thus choosing one unique element means we are choosing a basis for each point from a possibly infinite collection of basis element families (depending on X , we may not need Axiom of Choice, but the proof is for full generality, so we assume it). $\square$

Note 9.1. *An open cover (not rigorous definition, just for intuition sake) is a collection of subsets such that their union is equal to the whole space X . A basis acts like an "open cover for elements of the topology". Any open set in the topology defined by the initial definition wrt bases can be represented by the union of a subcollection of $\mathcal{B}$.*

In other words, that subcollection of basis elements covers that member of the topology so all that remains in the topology are the basis elements and their arbitrary unions and intersections.

Definition

Definition 9.14. We define the interior and closure:

1. The interior of a set A : $\text{int}(A) := \bigcup \{O \in \tau : O \subseteq A\}$
2. The closure of a set A : $\bar{A} := \bigcap \{C \in \tau : A \subseteq C\}$

The closure of the set relies on arbitrary intersections which is possible due to the arbitrary intersections of closed set collections in the topology.

Lemma

Lemma 9.15. $\text{int}(A) \subseteq A \subseteq \bar{A}$

Proof. If $x \in \text{int}(A)$, then:

$$\forall x : x \in \text{int}(A) \implies \exists O \subseteq A : x \in O$$

Since $x \in O \subseteq A$, then $x \in A$, so $\text{int}(A) \subseteq A$.

If $x \in A$, then:

$$\begin{aligned} \forall x, \forall C \in \tau : x \in A \wedge A \subseteq C &\implies x \in C \\ \implies \forall x : x \in A &\implies \forall C \in \tau : x \in C \iff x \in \bar{A} \end{aligned}$$

Therefore $A \subseteq \bar{A}$, proving our lemma. $\square$

Lemma

Lemma 9.16. The following equivalences hold:

1. A open $\iff A = \text{int}(A)$
2. A closed $\iff A = \bar{A}$

Proof. We prove the first statement:

$\implies$: If A is open, then A belongs to the union of open sets which are subsets of A , thus $A \subseteq \text{int}(A)$. We know that $\text{int}(A) \subseteq A$, thus $A = \text{int}(A)$.

$\impliedby$: If $A = \text{int}(A)$, then A is the union of open sets so A is an open set. $\square$

Proof. The proof of the second statement is similar:

$\implies$: If A is closed, then A belongs to collection of closed sets contributing to the intersection, thus $\bar{A} \subseteq A$. We know that $A \subseteq \bar{A}$, so $A = \bar{A}$.

$\impliedby$: If $A = \bar{A}$, then A is an arbitrary intersection of closed sets, which by topology axioms, A is closed. $\square$

Definition

Definition 9.17. The boundary of a set A or ∂A is defined as $\bar{A} \setminus \text{int}(A)$

If $A \subseteq X$, then we can write ∂A as $\bar{A} \cap (X \setminus \text{int}(A))$. Since both $\bar{A}$ and $X \setminus \text{int}(A)$ are closed, then their intersection is closed.

Lemma

Lemma 9.18. Let (X, ϱ) be a metric space. Prove that for all sets $A, B \subseteq X$, the following holds:

$$A \subseteq B \implies \text{int}(A) \subseteq \text{int}(B) \wedge \bar{A} \subseteq \bar{B}$$

and

$$X \setminus \text{int}(A) = \overline{X \setminus A} \wedge X \setminus \bar{A} = \text{int}(X \setminus A)$$

In particular.

$$\partial A = \bar{A} \cap \overline{X \setminus A} = \partial(X \setminus A)$$

Proof. We know that if $x \in A$, then $x \in \bar{A}$. Since $\bar{A} = \text{int}(A) \cup \partial A$, then $x \in \text{int}(A) \vee x \in \partial A$. If we consider $x \in \text{int}(A)$:

$$\begin{aligned} x \in \text{int}(A) &\iff \exists O \subseteq A : x \in O \\ \implies O \subseteq A \subseteq B &\implies O \subseteq \text{int}(B) \end{aligned}$$

We see the last implication as O is an open subset of B which means it contributes to the union of open sets that make up the interior. Thus $x \in \text{int}(B)$, showing $\text{int}(A) \subseteq \text{int}(B)$.

If $x \in A$ (not necessarily in the interior), then $x \in \bar{A}$:

$$x \in \bar{A} \implies \forall C \subseteq X : A \subseteq C \wedge C \text{ closed} \implies x \in C$$

Since $A \subseteq B$ and $B \subseteq \bar{B}$, then we see that $A \subseteq \bar{B}$. So $\bar{B}$ is one such set that contributes to the closure of A . Thus:

$$\forall x : x \in \bar{A} \implies x \in \bar{B} \iff \bar{A} \subseteq \bar{B} \quad (5)$$

We first prove the following statement:

$$A \subseteq B \implies X \setminus B \subseteq X \setminus A$$

If $x \in X \setminus B$ or B^c , then $x \notin B \implies x \notin A$ (contrapositive of inclusion), so $B^c \subseteq A^c$.

For the second statement, we see that by DeMorgan's Laws:

$$\begin{aligned} X \setminus \text{int}(A) &= X \setminus \bigcup \{O \subseteq X : O \text{ open} \wedge O \subseteq A\} = (\bigcup \{O \subseteq X : O \text{ open} \wedge O \subseteq A\})^c \\ &= \bigcap \{X \setminus O \subseteq X : X \setminus O \text{ closed} \wedge X \setminus A \subseteq X \setminus O\} \end{aligned}$$

Then by definition of closure, $X \setminus \text{int}(A) = \overline{X \setminus A}$. We now prove the statement for $X \setminus \bar{A}$:

$$\begin{aligned} X \setminus \bar{A} &= X \setminus \bigcap \{C \subseteq X : C \text{ closed} \wedge A \subseteq C\} = (\bigcap \{C \subseteq X : C \text{ closed} \wedge A \subseteq C\})^c \\ &= \bigcup \{X \setminus C \subseteq X : X \setminus C \text{ open} \wedge X \setminus C \subseteq X \setminus A\} = \text{int}(X \setminus A) \end{aligned}$$

Finally, by definition 9.6, we know that $\partial A = \bar{A} \cap (X \setminus \text{int}(A)) = \bar{A} \cap \overline{X \setminus A} = \overline{X \setminus A} \cap (X \setminus X \setminus \bar{A}) = \overline{X \setminus A} \cap (X \setminus \text{int}(X \setminus A)) = \partial(X \setminus A)$.

□

9.3 More Topological Notions

The following notions can be generalized for any topology by replacing the metric ball with any open set $U \in \tau$. However, since we verified that the open balls are a basis for a metric space topology and that every element in the topology is the union of basis elements/open balls, then we simply use the notion of open balls since they make up the topology τ_B :

Definition

Definition 9.19. An adherent point $x \in X$: $\forall r > 0 : B(x, r) \cap A \neq \emptyset$

In essence, an adherent point is a point for which we can find a subset in the topology intersection with A . Since our topology is constructed by open balls and their unions and intersections, such a set is an open ball.

Because of the notion of distance, we can make this open ball arbitrarily small.

Lemma

Lemma 9.20. $x \in X$ is adherent if and only if $x \in \bar{A}$

Proof. $\implies$: We prove the contrapositive:

$$\forall x \in X : x \notin \bar{A} \implies \exists r > 0 : B(x, r) \cap A = \emptyset$$

If $x \notin \bar{A}$, then $x \in X \setminus \bar{A}$. We prove the contrapositive: Since the complement is open and $A \subseteq \bar{A}$:

$$\exists r > 0 : B(x, r) \subseteq X \setminus \bar{A} \subseteq X \setminus A \implies B(x, r) \cap A = \emptyset$$

$\impliedby$: We prove by contrapositive again:

$$\forall x \in X : \exists r > 0 : B(x, r) \cap A = \emptyset \implies x \notin \bar{A}$$

Since $B(x, r) \cap A = \emptyset$, then $B(x, r) \subseteq X \setminus A$, so x is an interior point of $X \setminus A$. By Lemma 9.10:

$$x \in \text{int}(X \setminus A) \iff x \in X \setminus \bar{A}$$

Since $x \in X \setminus \bar{A}$, then we see that $x \notin \bar{A}$. □

We can conclude from this lemma that the set of adherent points is the closure of $A \subseteq X$.

Definition

Definition 9.21. A boundary point $x \in X$: $\forall r > 0 : B(x, r) \cap A \neq \emptyset \wedge B(x, r) \cap (X \setminus A) \neq \emptyset$

Since the boundary $\partial A := \bar{A} \setminus \text{int}(A) = \bar{A} \cap (X \setminus \text{int}(A))$. By the definition of boundary/adherent point, we see that:

$$x \text{ boundary pt} \implies x \in \bar{A} \wedge x \in \overline{X \setminus A}$$

By Lemma 9.10, we then see that $\overline{X \setminus A} = X \setminus \text{int}(A)$, so $x \in \partial A$. So the set of boundary points makes up ∂A .

Definition

Definition 9.22. An interior point $x \in X$: $\exists r > 0 : B(x, r) \subseteq A$. (Question to self, what if it were $\forall r > 0$?)

The set of interior points is the interior of the set A . This is because the open ball would belong to an open subset of A , so (will complete proof later)

Definition

Definition 9.23. An exterior point $x \in X$: $\exists r > 0 : B(x, r) \cap A = \emptyset$

The set of exterior points is the set $X \setminus \bar{A}$. (will complete proof later)

Definition

Definition 9.24. A limit point $x \in X$: $\forall r > 0 : (B(x, r) \cap A) \setminus \{x\} \neq \emptyset$.
An isolated point $x \in A$: $\exists r > 0 : B(x, r) \cap A = \{x\}$

Open balls around limit points still contain other points of A besides the limit point, unlike isolated points. So isolated points are the negation of limit points (rewritten as $\exists r > 0 : (B(x, r) \cap A) \setminus \{x\} = \emptyset$).

The idea of a limit point is similar to what was done with the ϵ proofs. We are just taking an arbitrary sized metric ball with radius ϵ or r , for which we must be able to find any point in that sequence (or in this case

space) within that radius.

Furthermore by definition, an isolated point always belongs to the set A (because it belongs in the intersection).

Lemma

Lemma 9.25. Every isolated point is a boundary point of A

Proof. We first tackle the easy requirement that an isolated point is adherent to A . Since the isolated point x is defined to be in A :

$$\exists r > 0 : B(x, r) \cap A = \{x\} \implies \forall r > 0 : B(x, r) \cap A \neq \emptyset$$

This is because $x \in A$ is always contained in any open ball radius r , so the intersection of the ball and A is always nonempty, thus x is adherent (thus its in the closure).

We now prove that $\forall r > 0 : B(x, r) \cap (X \setminus A) \neq \emptyset$ (it is adherent to the complement of the set). The definition of an adherent point tells us that $\exists r_0 > 0 : B(x, r) \cap A = \{x\}$. So:

$$r \leq r_0 \implies B(x, r) \subseteq B(x, r_0)$$

So since $x \in A$ and $x \in B(x, r)$, then $B(x, r) \cap A = \{x\}$. Thus all other elements in the open ball belong in $X \setminus A$.

If $r > r_0$, then $B(x, r_0) \subseteq B(x, r)$. Since $B(x, r_0)$ contains elements in $X \setminus A$, then so does $B(x, r)$. So we have shown for any isolated point $x \in A$:

$$\forall r > 0 : (B(x, r) \cap A \neq \emptyset) \wedge (B(x, r) \cap (X \setminus A) \neq \emptyset)$$

□

Lemma

Lemma 9.26. Every limit point is adherent.

Since a limit point $x \in X$ is defined such that $\forall r > 0 : (B(x, r) \cap A) \setminus \{x\} \neq \emptyset$, then we see that $B(x, r) \cap A \neq \emptyset$, thus it is adherent or in the closure of A .

Note that the limit of a sequence is a limit point or isolated point, but the converse is not necessarily true. So in following proofs, we use the stated implication.

Lemma

Lemma 9.27. $\bar{A} = \{x \in X : \text{limit point}\} \cup \{x \in X : \text{isolated point}\}$ where the two sets in the union are disjoint.

Proof. The disjointness of the two sets follows from the definitions. In Defn 9.12, we rewrote the statement of an isolated point which is the opposite of a limit point, so a point cannot be both a limit point and isolated point.

$$x \in \bar{A} \iff \forall r > 0 : B(x, r) \cap A \neq \emptyset$$

We know that a point x in the closure is either an isolated (and adherent) or adherent but not isolated, so:

$$\iff (\exists r > 0 : B(x, r) \cap A = \{x\}) \vee ((\forall r > 0 : B(x, r) \cap A \neq \emptyset) \wedge \neg(\exists r > 0 : B(x, r) \cap A = \{x\}))$$

$$\iff (\exists r > 0 : B(x, r) \cap A = \{x\}) \vee ((\forall r > 0 : B(x, r) \cap A \neq \emptyset) \wedge \forall r > 0 : B(x, r) \cap A \neq \{x\})$$

The expression simplifies to the point being isolated or a limit point, a different definition for closure than interior and boundary union. Since the closure is the set of adherent points, we see every adherent point is either a limit point or isolated point. $\square$

9.4 Sequential Closedness

Definition

Definition 9.28. A first countable space is one for which each point has a countable local basis.

Recall how we defined a basis for a topology. Now a local basis is defined as a basis for all open sets that contain the point x . To define this, we require a pre-existing topology, for which we can then use a basis element of that topology.

The previous notions in 9.3 were purely topological given a basis of open balls. First countability allows us to equate them to sequential statements.

Lemma

Lemma 9.29. All metric spaces are first countable.

Since we are working in metric spaces, we will use the assumption that all metric spaces are first countable.

Proof. $\square$

Lemma

Lemma 9.30. $(\exists \{x_n\}_{n \in \mathbb{N}} \in A^{\mathbb{N}} : x_n \rightarrow x, \forall n \in \mathbb{N} : x_n \neq x \iff x \text{ limit point}$

This lemma says that if there exists a convergent A -valued sequence such that no term equals the limit of the sequence, then the limit is a limit point.

Proof. $(\implies)$: Since it converges to x , we see that $\forall r > 0, \exists n_0 \geq 0, \forall n \geq n_0 : x_n \in B(x, r) \wedge x_n \neq x$, so:

$$\forall r > 0 : (B(x, r) \cap A) \setminus \{x\} \neq \emptyset$$

Thus x is a limit point by definition. We now prove the converse: $(\impliedby)$: If x is a limit point of the set A , then by definition:

$$\forall r > 0 : (B(x, r) \cap A) \setminus \{x\} \neq \emptyset$$

We use the idea of first countability here to construct a local basis. We then use Axiom of Choice to choose a sequence (refer to Thm 9.1). $\square$

Lemma

Lemma 9.31. $(\exists \{x_n\}_{n \in \mathbb{N}} \in A^{\mathbb{N}} : x_n \rightarrow x, \exists n_0 \in \mathbb{N}, \forall n \geq n_0 : x_n = x \iff x \text{ isolated point}$

Proof. $(\implies)$: We prove by contrapositive here. If x is a limit point, then $\forall r > 0 : (B(x, r) \cap A) \setminus \{x\} \neq \emptyset$. Using first countability of a metric space and the Axiom of Choice construction in 9.1, we can find a sequence where $\forall n \in \mathbb{N} : x_n \neq x$.

$(\impliedby)$: If x is an isolated point, then there must exist exist an eventually constant sequence. Here we do not require first countability or axiom of choice, we simply take the constant sequence $\{x\}_{n \in \mathbb{N}}$. $\square$

Theorem

Theorem 9.32. $A \text{ closed} \iff (\forall \{x_n\}_{n \in \mathbb{N}} \in A^{\mathbb{N}}, \forall x \in X : x_n \rightarrow x \implies x \in A)$

Proof. We prove both implications:

($\implies$) For a sequence $\{x_n\}$, because A is closed, then $A = \bar{A}$. If $x_n \rightarrow x$, then by the definition of convergent:

$$\forall r > 0, \exists n_0 \geq 0, \forall n \geq n_0 : x_n \in B(x, r)$$

Since $x_n \in A$, then $B(x, r) \cap A \neq \emptyset$, thus x is an adherent point so $x \in A = \bar{A}$.

Note: Alternatively A closed means $A = \bar{A}$. Then for all convergent sequences, the limit is either an isolated point or limit point by 9.16/9.17. Thus by 9.14 all limits are in $\bar{A} = A$.

($\impliedby$) We prove by contrapositive. If A is not closed, then we show that $x_n \rightarrow x \wedge x \notin A$ is true. If $x \notin A$ then $x \in X \setminus A$ and $X \setminus A$ is not open since A not closed.

Therefore, (informally), $X \setminus A$ has a boundary part (its "closen"), so we see that:

$$\forall r > 0 : B(x, r) \cap A \neq \emptyset$$

Since $\forall x_n : x_n \in A$, we need to find a sequence such that $\forall \epsilon > 0, \exists n_0 \geq 0, \forall n \geq n_0 : \varrho(x_n, x) < \epsilon$. Alternatively $\forall \epsilon > 0, \exists n_0 \geq 0, \forall n \geq n_0 : x_n \in B(x, \epsilon)$. (This is the same as the proposition of an open ball is $\varrho(x, x_n) < \epsilon$)

(Axiom of Choice): The sequence allows us to say that $B(x, r) \cap A \neq \emptyset$ as $\forall n \geq 0 : x_n \in A$. To choose from an infinite family, we require the axiom of choice. We give the choice rule and because $x \notin A$:

$$\{x_n\}_{n \in \mathbb{N}} := \bigtimes_{n \in \mathbb{N}} A \cap (B(x, 2^{-n}) \setminus \{x\}) = \bigtimes_{n \in \mathbb{N}} (A \cap B(x, 2^{-n}) \setminus \{x\})$$

Inductively we see that we create an A -valued sequence such that we always get that $x_n \in B(x, 2^{-n})$ or that $\varrho(x_n, x) < 2^{-n}$ ensuring convergence while $x \notin A$. We see that we use the definition of x being a limit point of the set A and equating it to the limit of a sequence (which under the hood relies the Hausdorff condition). $\square$

Corollary

Corollary 9.33. If $A \subseteq X$, then:

$$\forall x \in X : x \in \bar{A} \iff \exists \{x_n\}_{n \in \mathbb{N}} \in A^{\mathbb{N}} : x_n \rightarrow x.$$

Proof. ($\implies$) If $x \in \bar{A}$, then it is adherent so $B(x, r) \cap A \neq \emptyset$. To choose a sequence such that the intersection is true $x_n \in B(x, r) \cap A$, we use the same process as Thm 9.1 to obtain a sequence (requires Axiom of Choice; this construction has to do with first countability of metric spaces; this is the contrapositive of Lemma 9.17) If x is an isolated point, then by contrapositive of Lemma 9.16, we can find a sequence (which is the constant sequence).

($\impliedby$) If we can find a sequence that converges to x , then $\forall r > 0, \exists n_0 \geq 0, \forall n \geq n_0 : x_n \in B(x, r)$. The point x is then either a limit point or an isolated point depending on the behavior of the sequence, so $x \in \bar{A}$. $\square$

Instead of talking only about adherent points, we can utilize Lemma 9.16 and Lemma 9.17 by categorizing x as a limit point or isolated point by Lemma 9.14 (A closure is disjoint union of limit points and isolated points).

Corollary

Corollary 9.34. Thm 9.1 simplifies to A is closed if and only if it includes all of its limit points.

Since the limits of sequences are either limit points or adherent points, and by definition adherent points are in $A \subseteq \bar{A}$, then we care about the A -valued sequences that converge to limit points of the set only.

Definition

Definition 9.35. A set $B \subseteq A$ is dense in A iff $\forall a \in A, \exists \{b_n\}_{n \in \mathbb{N}} \in B^{\mathbb{N}} : b_n \rightarrow a$

We see by the previous lemma that if we have a b -valued convergent sequence, then it must converge to something in the closure of B as the limit point is either in the interior or boundary.

Since we already define all points in A as the limit points of b -valued convergent sequences, then we see that all points in A must belong to the closure of B so $A \subseteq \bar{B}$. This is an equivalent definition for (sequentially) dense.

We can build on this further. If we have a metric, then we have a topology generated by the basis of open balls. Then by the definition of convergent sequences:

$$\forall r > 0, \exists n_0 \geq 0, \forall n \geq n_0 : b_n \in B(a, r)$$

So we can always find an arbitrarily close point of the sequence to a , so no matter how close we look (eg if we look at some interval of points in A centered around a then we can always find some point in B).

Definition

Definition 9.36. A metric space is deemed **separable** if:

$$\exists A \subseteq X : A \text{ countable} \wedge \bar{A} = X$$

The reals $\mathbb{R}$ for example are separable as we can find a countable subset (e.g. the rationals $\mathbb{Q}$) such that their closure is the reals themselves. When we say $\bar{A} = X$, we mean that A is dense in X .

This is due to our equivalent definition of density. If A is dense in X , then $X \subseteq \bar{A}$ since for all $x \in X$, we can find an A -valued sequence whose limit point is x , thus $x \in \bar{A}$. However, since X is the whole space it is not possible that $X \subset \bar{A}$, thus $X = \bar{A}$.

So a metric space X is separable if:

$$\exists A \subseteq X : A \text{ countable} \wedge A \text{ dense}$$

10 Metric Completeness

10.1 Key Statements and Definition of Completeness

Definition

Definition 10.1. A metric space (X, ϱ) is complete if all Cauchy sequences converge with a limit point in that space:

$$\forall \{x_n\}_{n \in \mathbb{N}}, \exists x \in X : \{x_n\} \text{ Cauchy} \implies x_n \rightarrow x$$

Definition

Definition 10.2. A set $A \subseteq X$ for (X, ϱ) is said to be bounded if:

$$\exists x \in X, \exists r > 0 : A \subseteq B(x, r) \quad (6)$$

By this definition, every open ball is bounded (by itself). We see the following connection between bounded sets and Cauchy sequences:

Lemma

Lemma 10.3. $\forall \{x_n\}_{n \in \mathbb{N}} : \{x_n\} \text{ Cauchy} \implies \{x_n : n \in \mathbb{N}\} \text{ bounded}$

Proof. If the sequence is Cauchy then we have the following definition:

$$\forall \epsilon > 0, \exists n_0 \geq 0, \forall m, n \geq n_0 : \varrho(x_m, x_n) < \epsilon$$

We let $n = n_0$ and choose $\epsilon = \epsilon_0$. Then by the triangle inequality for an $x \in X$:

$$\varrho(x, x_m) \leq \varrho(x, x_n) + \varrho(x_n, x_m) < \varrho(x, x_n) + \epsilon_0$$

The left hand side covers all sequence members (for all naturals). The right hand side does as well, however it splits it into two sets. The first $\varrho(x, x_n)$ can act on $n < n_0$ and the second can act on $n \geq n_0$.

We let $r := \max\{\varrho(x, x_k) : k \leq n\}$, so that $\forall x \in X : \varrho(x, x_m) < r + \epsilon_0$, which means, we have a ball $B(x, r + \epsilon_0)$ such that the set of all terms of the sequence belong to this metric ball. $\square$

Lemma

Lemma 10.4. Given a totally ordered set $(A, \leq)$: for each A -valued sequence, there exists a monotonically increasing (non-decreasing so $\leq$) or decreasing subsequence.

I will complete the proof later.

Lemma

Lemma 10.5. Every bounded monotone sequence of elements from an ordered, complete field converges such that:

$$\lim_{n \rightarrow \infty} x_n = L := \sup\{x_n : n \in \mathbb{N}\}$$

OR if it is non-increasing:

$$\lim_{n \rightarrow \infty} x_n = L := \inf\{x_n : n \in \mathbb{N}\}$$

Proof. We let a monotone sequence either be non-increasing or non-decreasing; since the set of elements is bounded, by completeness of F , it admits suprema/infima. Then we show that the supremum of the terms of the sequence serves as a limit:

$$\forall \epsilon > 0 : L - \epsilon = L \implies \exists x_{n_0} : L - \epsilon < x_{n_0}$$

This implication comes from the fact that the if $L - \epsilon$ were an upper bound then $\forall n \in \mathbb{N} : x_n \leq L - \epsilon$. The existence of the specific x_n is the negation of this statement. We see further that by the monotonicity that $\forall n \geq n_0$:

$$L - \epsilon < x_{n_0} \leq x_n < L + \epsilon$$

OR if its non-increasing:

$$L - \epsilon < x_n \leq x_{n_0} < L + \epsilon$$

Then we manipulate both inequalities and get that $\forall n \geq n_0 : |x_n - L| < \epsilon$, thus matching the definition of the limit of a sequence. $\square$

Theorem

Theorem 10.6. Bolzano-Weierstrass: Every bounded sequence in a complete, ordered field has a convergent subsequence.

Proof. The proof for the theorem is now simple and completed by the previous two lemmas. Since the sequence is bounded, any subsequence is merely a subset of the elements of the sequence, which by transitivity of set inclusions is also bounded. By Lemma 10.2, we can find a monotonic subsequence and by Lemma 10.3, we know that it converges. $\square$

Corollary

Corollary 10.7. $\mathbb{R}$ is then complete.

Proof. Given any Cauchy sequence in $\mathbb{R}$, by Lemma 10.1 it is bounded. Since it is bounded, we can find a convergent subsequence by Bolzano Weierstrass. By Lemma 8.3, since the original sequence was Cauchy, it converges. $\square$

This theorem can be extended to any n dimensional real space with the Euclidean metric. I will add more details later.

10.2 Completeness Related to Closedness

We offer a brief summary of the properties listed so far and make a distinction between purely topological properties and purely sequential properties:

Defined by Metrics :=	Topological	Sequential
	Open sets Closure/closed sets	Cauchy/convergent sequences Completeness

The notion of completeness is purely sequential (all cauchy sequences converge to a point in the space), which by our definition, requires a metric. It is possible without a metric (adding something called a uniform structure), but it is not possible in a regular topological space.

A regular topological space only gives us notions of closedness/openness and not of distance, thus we cannot define "arbitrarily close" points. When we have a metric, we can define open sets by a basis of open balls and talk purely in the topological sense:

1. A metric gives us a basis
2. A basis by Lemma 9.6 generates a topology and defines open sets (which by Lemma 9.7 are covered by basis elements)
3. Closed sets are then defined as one whose complement is open

However, here we only discuss the notions in a purely topological sense. They lead to further definitions such as adherent points, limit points, etc, but the idea of a closure becomes quite clunky to define.

Recall how we defined the closure as the intersection of all closed supersets. However, we can make use of the metric to construct convergent sequences to define the limits of sequences as limit points or isolated points, thus accounting for the whole (topological) closure defined by notions of adherent/limit/isolated points.

We defined the notions of sequential closure which led to density of a set in another set; because we have that metric spaces are first countable we are able to say sequential closure is equivalent to topological closure by the methods listed in the previous paragraph.

Just for more detail, completeness and sequential closure will be notated with respect to the metric for example:

(A, ϱ) complete $:= A$ is a complete metric space/subspace

(A, ϱ) closed $:= A$ is sequentially closed

We know sequentially closed is equivalent to topologically closed, but we will continue to differentiate the notions in proofs. Closed with a metric means sequentially as a metric is a necessary structure for convergence, and closed without a metric written means topologically (though there still may be an underlying metric generating a topology).

Now completeness was purely a property of sequences in a space, but the following lemma shows the topological relation using closedness as the bridge:

Lemma

Lemma 10.8. Given a complete metric space (X, ϱ) , for all $A \subseteq X$:

$$(A, \varrho_A) \text{ complete} \iff A \text{ closed}$$

Proof. ($\implies$): If A is complete, then all Cauchy sequences converge to some point in A . Thus convergent sequences converging to a point in the set means it is sequentially closed. By Thm 9.1, it is then topologically closed.

($\impliedby$): If A is not complete, then (will finish later). □

Note that the "induced metric" ϱ_A . The topology generated by this metric is called the subspace topology, but we can find that the two topologies of X and A are equal, so a sequence in A has the same properties in X (not necessarily the other way around).